

Chapter 2

Paper I. Political Deliberation as Network: An Applied Methodological Note

2.1 Introduction

Political deliberation as both a collective decision-making and an opinion-sharing mechanism has been an important topic for analysts and practitioners alike ([Bächtiger et al. 2018](#); [Myers and Mendelberg 2013](#)). Many have considered its normative and empirical properties ranging from facilitating de-polarization and promoting decision legitimacy to improving policy quality and encouraging minority participation.¹ In comparison, its methodological treatment has received less attention, with most empirical analysis employing simple descriptive statistics and conventional regression methods² which are insufficient to handle the multidimensional (at individual, group, and discourse levels) and unit-codependent nature of the deliberative process ([Beauchamp 2020](#); [Miller and Sutherland 2022](#)). Such naive approaches are known to yield more biased results and lead to less reliable inference in situations involving dyadic interdependence and context effects. From the perspective of the applied researcher, this would likely lead to false impressions with inflated confidence about the true information contained by the data and harm the quality of substantive inquiry based on them.

¹See e.g., [Austin-Smith and Feddersen \(2006\)](#), [Humphreys, Masters and Sandbu \(2006\)](#), [Mutz \(2008\)](#), [Patty and Penn \(2011\)](#), [Collins \(2021\)](#), [Fishkin et al. \(2021\)](#), [Groenendyk and Krupnikov \(2021\)](#), and many others.

²To put things in context with a simple publication metric, for all published political science papers on deliberation available on Google Scholar, only about 120 publications include the word “ERGM” (the main type of network model focused on in the following sections) and over 8,000 mention the word “logit” or “logistic” (the main type of conventional regression models that is the closest analogue to the ERGM. It is clear that traditional regression models have been on an order of magnitude more popular than network models.

In this paper a network approach is proposed as a better alternative for analyzing the deliberative processes. It proposes to measure the process as a graph (network) with the participants as vertices (nodes) and their conversational interactions as edges (links), and apply statistical models for network inference to study effects of interest. The two key benefits of the network approach are a better respect for the complex structure of the deliberative process and an explicit accounting of the social dynamics like homophily or reciprocity that are both factor and outcome of the same process. In particular, when the latter are considered as key estimands under inquiry, by definition only network models may possibly facilitate their statistical estimation. The paper focuses on the most popular class of network models, the Exponential Random Graph Model (ERGM) family, and demonstrates the methodological utility of the network approach on two real-life datasets with different deliberative activities. The network-ERGM approach is relatively simple to understand and implement and yields superior insights into the complex generative process driving deliberation than simple regression approaches. Below proceeds in the order of theoretical reasoning, model description, empirical demonstration, and a brief discussion at the end.³

In terms of contribution, the paper aims more to make a principled case for the *generalized* use of network models (ERGM or others) for the empirical study of political deliberation as the default modelling approach, rather than simply applying the model to certain cases and interpret the results for specific theoretical questions as such. Several important contributions already exist in this regard (notable examples include, for instance: [Song 2014](#); [Gonzalez-Bailon, Kaltenbrunner and Banchs 2010](#); [Sobkowicz, Kaschesky and Bouchard 2012](#); [Kim and Park 2012](#)). The goal is to push such models to the forefront of empirical application to improve the average quality of statistical inference in the said field of study, from a broader applied methodological perspective, not specific research designs or findings in a narrower sense. As such, the case made in this paper is more for a different methodological approach rather than for different model specifications within a conventional method. The case demonstrations used below reflect this general concern.

It should further be noted that the modelling approach proposed in this paper extends beyond strictly deliberative contexts, and in principle applies to any context in which network structures can be identified and correspondingly data be collected. Common examples that immediately come to mind include, for instance, public opinion networks with key roles of the opinion leaders; conflict networks among international and domestic actors; issue competition in multidimensional policy spaces between political parties; and much more. Deliberation is a special case with which the network approach affiliates itself most naturally, in this respect.

³Supporting information can be found in the attached appendix.

2.2 Why Network Deliberation?

Modelling deliberation as a network is both a natural and a profitable approach for political scientists. The “natural” part refers to the fact that each important component in the former’s conceptual space may find a unique representation in the latter’s mathematical space; and the “profitable part” refers to the fact that by doing so we gain richer and more subtle insights into the complex effect dynamics and mechanisms at play in shaping the deliberative process. Under a usual set of assumptions the workhorse model for network analysis, the ERGM, may recover the effects of both exogenous attributes of the participants and endogenous features of the deliberative setting in a relatively easy way.

As table 2.1 summarizes, there is a simple one-to-one mapping between deliberation as an empirical process and network as a mathematical construct, and the ERGM (or more complex models) as a statistical instrument: from a deliberative process consisting of its participants and their interactions, to a graph consisting of the vertex (V) and edge (E) sets; from the relationship between the participants defined by their interactions, to the adjacency matrix X with row and column dimensions equal to the total number of participants; from the tendency of these participants to interact with each other, to the probability of forming ties between vertices ($P(X|\theta)$); and from the effects of various independent and self-dependent factors upon such interactive tendencies, to the coefficient-weighted per-unit change in the corresponding nodal covariates ($\theta_{\text{exo}}^T \sigma [s_{\text{exo}}(X)]$) and network statistics ($\theta_{\text{endo}}^T \sigma [s_{\text{endo}}(X)]$). In the two lattermost cases, the θ ’s represent the effect vectors of interest, $s(X)$ ’s represent the nodal covariates on the graph, and the σ ’s the per-unit change in such covariates, such that the entire expression is the effect of a unit change in each covariate associated with the nodes on the graph-level outcome. The graph may also assume a more complex character, such as a directed graph when the interaction takes a more directional attribute like interruptions or addresses, or a weighted one when the interactions are counted in more than binary terms. Regardless, the network model allows for a natural and elegant way to represent the multilayered and multidimensional nature of the deliberative process.

Such clear match between the substantive (deliberation) and technical (network) studies brings two benefits for analysts in making empirical inference. First and as a direct result of the non-tabular representation, one can easily draw inference at different levels of the deliberative process by calling on the appropriate descriptive statistics, from the individual-level distribution of sociodemographic attributes to the group-level summary of interaction types. The need for overly complicated transformation to accommodate such complex data structure is thus reduced by construction and the inferential process more streamlined and efficient. The empirical applications below (section 2.4) will show how simple network statistics

Table 2.1: The methodological mapping under the current approach.

Deliberation	Network	ERGM
Participants	Vertices	V
Interactions	Edges	E
Relations	Adjacency matrix	X
Process	Graph	$G(V, E)$
Propensity to interact	Probability of forming ties	$P(X \theta) = \frac{1}{\kappa(\theta)} \exp \{ \theta^T s(X) \}$
Exogenous dynamics	Effects of exogeneous covariates	$\theta_{\text{exo}}^T \sigma [s_{\text{exo}}(X)]$
Endogeneous dynamics	Effects of endogenous measures	$\theta_{\text{endo}}^T \sigma [s_{\text{endo}}(X)]$

like degree distribution or triad census help reveal the systematic patterns of uneven interaction among participants in two settings.

Second and in relation to conventional regression approaches, the network approach allows for the explicit estimation of network-endogenous effects whose determinants entail dependency between units, such that common assumptions of SUTVA or i.i.d. sample need not apply or be invoked on untenable grounds. As will be made clear in section 2.3, the likelihood of one participant interacting with another may depend not only on their own characteristics external to the process, such as age or gender, but also certain features internal to the very process itself, be it the prior experience of the latter interacting with this person first (reciprocity) or both having interacted with a third participant beforehand (transitivity). This explicit modelling of conditional tie-forming likelihood as a function of both exogenous and endogenous factors is particularly helpful for analyzing deliberative processes, where dyadic interactions between participants are rarely independent from each other, due both to the mechanical effect of having the same individual involved in multiple interactions and to the social effect of having all individuals enclosed in a relatively closed and mutually aware setting (Austen-Smith and Feddersen 2006; Miller and Sutherland 2022). Results from empirical analysis also support the non-trivial significance of network effects on deliberative processes in the two settings: without controlling for such effects the estimated influence from exogenous factors are very likely inflated and the model fit to the observed structure unsurprisingly worse.

2.3 The Model

The ERGM family are one of the most widely used family of models for network inference today. Under this class of models, the conditional probability of observing a given network x

is modeled as the realization of random graph X using an exponential family link ([Schmidt and Morup 2013](#); [Krivitsky 2012](#)):

$$\begin{aligned} P(X = x|\theta) &= \frac{1}{\kappa(\theta, \mathcal{X})} \exp \{ \theta^T s(x) \} \\ &= \frac{1}{\kappa(\theta, \mathcal{X})} \exp \{ \theta_{\text{exo}}^T s_{\text{exo}}(x) + \theta_{\text{endo}}^T s_{\text{endo}}(x) \}, \end{aligned} \quad (2.1)$$

where θ is the vector of coefficients, $s(x)$ a vector of sufficient statistics, and $\kappa(\theta)$ the normalizing constant over the graph space $\mathcal{X}$ ensuring the distribution is proper. For the present interest, $s(x)$ can be further decomposed into two types, the exogenous predictors whose realization do not depend on the network ($s_{\text{exo}}(x)$) and the endogenous statistics who are computed on the network itself and induce dependency between dyads ($s_{\text{endo}}(x)$). When a model includes only exogenous terms it is called a dyad independent model, otherwise it is called dyad dependent. $\kappa(\theta)$ is often intractable as it involves summing over all possible networks so is usually approximated via Markov chain Monte Carlo (MCMC) sampling instead ([Duxbury 2021](#)).

Equation (2.1) can be rewritten to express the conditional probability of individual ties ([Harris 2014](#)), where the response is shifted from the entire graph (X) to each dyad within the graph (X_{ij}) under a conditional independence assumption between each chosen dyad and the rest of the dyads in the network:

$$\begin{aligned} P(X_{ij} = 1 | n_V, X_{ij}^C) &= \text{logit}^{-1} \left\{ \sum_{k=1}^K \theta_k \sigma[s_k(x)] \right\} \\ &= \text{logit}^{-1} \left\{ \sum_{k_{\text{exo}}=1}^{K_{\text{exo}}} \theta_{k_{\text{exo}}} \sigma[s_{k_{\text{exo}}}(x)] + \sum_{k_{\text{endo}}=0}^{K_{\text{endo}}} \theta_{k_{\text{endo}}} \sigma[s_{k_{\text{endo}}}(x)] \right\}, \end{aligned} \quad (2.2)$$

where $\sigma[s(x)]$ is the change statistic representing the change in a network statistic when X_{ij} goes from 0 to 1. Following (2.1), the network statistics can again be decomposed into an exogenous and an endogenous subset with their respective coefficient terms. n_V is the total number of vertices in the network and X_{ij}^C all the dyads in the network other than X_{ij} . Equation (2.2) makes it clear that the probability of each tie is dependent on the rest of the network. For instance, when the change statistic refers to the total number of links in the observed network, the probability of each pair of nodes forming a tie is modelled to be dependent on this network-level attribute. When the associated effect parameter assumes a

positive (negative) value, the model would suggest that pairwise link probability is positively (negatively) affected by the total number of existing ties in the entire network.

Following [Cranmer and Desmarais \(2011\)](#), it is useful to note the special case when $K_{endo} = 0$, i.e., the model includes only dyad-independent predictors:

$$P(X_{ij} = 1 | n_V, X_{ij}^C) = \text{logit}^{-1} \left\{ \theta_E + \sum_{k_{\text{exo}}=0}^{K_{\text{exo}}} \theta_{k_{\text{exo}}} \sigma[s_{k_{\text{exo}}}(x)] \right\}, \quad (2.3)$$

which reduces to a binary logistic regression where the edge count term (whose change statistic is always 1) θ_E is the intercept and other change statistics the predictor terms.⁴ This equivalence makes the ERGM a more legible model for practitioners more familiar with logistic regressions. Empirical applications below will show that this is indeed the case: a dyad-independent ERGM yields the same parameter estimates as a logit model on the full dyad set in real-life data.

To save space, issues related to model estimation and limits are left to appendix section A.1. The two important assumptions are however worth pointing out here ([Cranmer and Desmarais 2011](#)):

- Each network statistics calculated on the observed graph X is the expected value of those statistics across all possible graphs;
- Only the network statistics included in the model affect the probability of observing X .

with the first assumption amounting to a necessity condition allowing inference to be made on observed data and the second a sufficiency condition allowing the model to be identified. To be more detailed, the first assumption requires that the average value of the network statistics over all the possible realizations of the underlying random graph should be no different from the ones computed on the given one (akin to an unbiased measurement requirement), and the second assumption requires that the model has been properly specified such that no relevant statistics have been left out (akin to a no-confounding requirement). These are no stronger assumptions than those commonly used in observational inference.

A more practical objection against the use of ERGM concerns the issue of model degeneracy, in which the fitted model yields unrealistic graphs that are either fully connected or empty ([Hunter et al. 2008](#); [Cranmer and Desmarais 2011](#)). Whilst a variety of tools exist on post-estimation checks against the issue, as [Cranmer and Desmarais \(2011\)](#) have pointed out, model degeneracy itself can act as a useful diagnostic mechanism that guards against poor

⁴Note that the index for exogenous predictors, k_{exo} can now start from zero as we have at least one non-trivial term in the model (the edge-count intercept).

specification and highlights the need for proper calibration, since poorly fitted models are more likely to generate poorly fitting graphs. In addition, one could always resort to other families of network models for cross-validation between model results or more stable estimation.

For analyzing deliberative processes, the ERGM therefore offers a relatively simple, feasible, and legible way for modelling its complex effect mechanisms, most especially those arising from network dynamics themselves, and thus deliver more reliable insights than conventional regression approaches. The next section will demonstrate such gains with two real-life datasets.

2.4 Applications

The usefulness of the endorsed approach is demonstrated with two publicly available datasets of political deliberation. The first dataset records online interactions (mentions) among Dutch MPs on Twitter ([Del Valle, Broersma and Ponsioen 2022](#)), and the second dataset records in-person interactions (interruptions) among US Congress members in committee hearings ([Miller and Sutherland 2022](#)). For empirical relevance, the focus is on gender inequality in the deliberative process, a key issue increasingly under investigation and supported with evidence (see e.g., [Mendelberg, Karpowitz and Oliphant 2014](#); [Mendelberg, Karpowitz and Goedert 2014](#)). Whether men and women are treated differently due to their gender identities is an issue of genuine public and policy concerns.⁵ The analysis here reconsiders the issue by treating each dataset as a directed graph and fitting ERGM of the following form:

$$P(X_{ij}^m = 1 | n_V^m, (X_{ij}^m)^C) = \text{logit}^{-1} \left\{ \theta_E^m + \sum_{d=1}^D \theta_d^m [\text{GenderPair}]_d^m + \sum_{k_{\text{exo}}^m=1}^{K_{\text{exo}}^m} \theta_{k_{\text{exo}}}^m \delta_{\text{exo}}^m + \sum_{k_{\text{endo}}^m=0}^{K_{\text{endo}}^m} \theta_{k_{\text{endo}}}^m \delta_{\text{endo}}^m \right\}, \quad (2.4)$$

where m indexes for the dataset, d indexes for four possible gender pairings with one left as the reference category,⁶ and δ a shorthand for the change statistics (control terms) mentioned before. θ is the parameter of interest whose magnitude and statistical significance serve as indicators for the differences in interaction probabilities between different gender dyads. The effects of particular interest are the θ_d 's which quantify the degree to which gender pairing (GenderPair_d) matters for driving deliberative activities among participants in

⁵Unfortunately, neither dataset defines gender in a non-binary way. The present analysis thus follows the binary scheme for practical reasons.

⁶ $\text{GenderPair} \in \{\text{Men to Men, Men to Women, Women to Men, Women to Women}\}$; Men to Men is the reference category in all analyses.

each case. Two ERGMs are fitted and compared against each other: a dyad-independent one without endogenous terms, and a dyad-dependent one with at least one endogenous term.⁷ All models are fitted with MCMC maximum likelihood estimation using the `ergm` package in R.

To preview, the models detect clear signs of gender bias – women are less likely to be mentioned on Twitter by their male than female colleagues – in the Dutch case but no similar patterns about person-to-person interruptions in the American case. Further, such effects are substantially depressed when network endogenous effects are accounted for. That is, the models ignoring network endogenous effects (in effect, conventional logistic regression models) likely returned overly optimistic estimates about the influence of gender effects. The sections below report each analysis in greater detail and, if taken at face value, could lead to overconfident conclusions about gender effects in deliberative settings.⁸ To simplify notations, whenever directed interaction of one group to another is concerned, the symbolic expression $A \rightarrow B$ is adopted, where A is the sender of the action-tie and B is the receiver of such action-tie. so e.g., $M \rightarrow W$ in the context of interruptions refers to “a [M]ale participant interrupting a [F]emale participant”.

2.5 Case I: Dutch Parliamentary Twitter Mentions

The Dutch data come from [Del Valle, Broersma and Ponsioen \(2022\)](#) and include directed interactions between 144 MPs on Twitter in a one-year window (November 2015 - November 2016). Each MP is defined as having a directed tie to another if she/he mentioned that person on Twitter during this period. For a detailed explanation on using Twitter mentions as a reasonable measure for online deliberation, see [Del Valle et al. \(2020\)](#). For a detailed description of the data and variables, see attached appendix sections A.2.1 and A.3.1.

Results from models in (2.4)’s form are summarized in figure 2.1. The left panel plots the average marginal effects (AMEs) of the three non-reference gender pairs on the probability of

⁷To be clear, both studies have already collected edge-level data and have used ERGM models in their empirical analyses. These are relatively rare among empirical studies of deliberation, as noted before, and the goal of the present analysis is not to replicate or contradict what the respective author teams have already done in theoretically rigorous ways. The present goal is rather to simply use the data the authors have made available to demonstrate to the applied audience how ERGM models can be used and the models results interpreted to yield improved insights about the process under study compared to conventional regressions. As such the models fitted below are simple and straightforward (e.g., without covariate interactions) and meant to be seen in a demonstrative light only.

⁸This is not to be taken as a catch-all statement that denies the presence of gender bias in deliberative settings. It only serves to emphasize the inflated effect size and precision estimates given by logit estimates in the case of the two utilized datasets here.

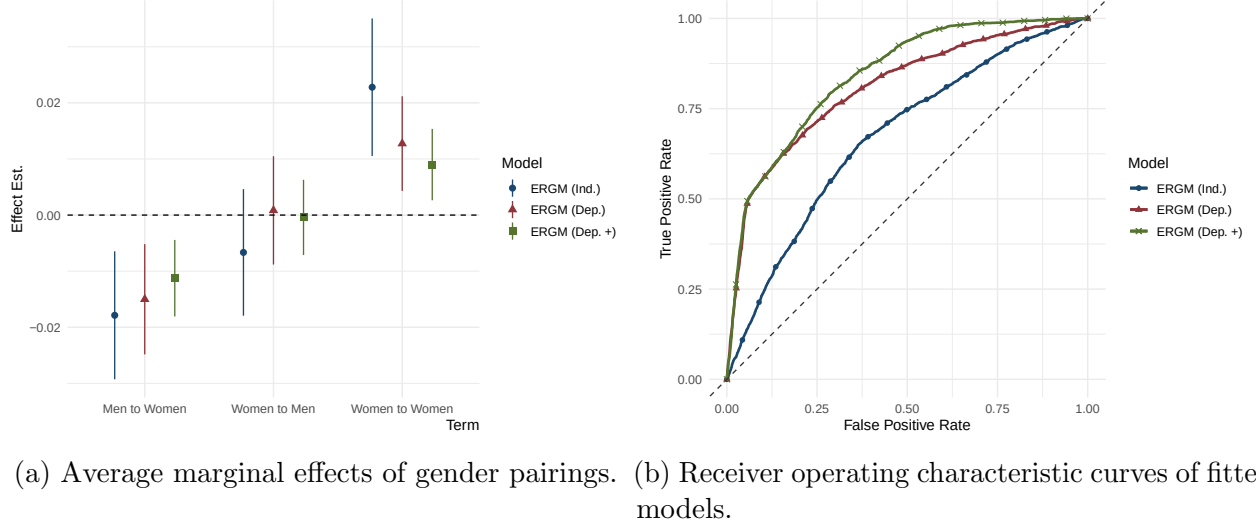


Figure 2.1: Summary of model results for the Dutch parliamentary Twitter mentions data.

being mentioned on Twitter, conditioning upon a set of dyad-interactive controls⁹; the right panel plots the receiver operating characteristics (ROC) curves showing the comparative fit of the models with observed data. We see that relative to the all-male baseline, women are indeed less likely than men to receive a Twitter mention ($\widehat{AME}_{M \rightarrow W} \approx [-2\%, -1\%]$), but are more likely to receive one from their fellow women ($\widehat{AME}_{W \rightarrow W} \approx [+1\%, +2\%]$).¹⁰ Meanwhile no significant difference statistical or substantive has been found between mixed-gender pairings. These results decompose and add considerable nuance to the authors' original findings using simpler ERGM specifications that women are overall more likely to be mentioned online. Notably, the models including network-endogenous terms – the red one including a reciprocal mentions and the green one including this and two additional terms capturing transitive and cyclic mentions in the figure¹¹ – both return smaller coefficient estimates (left panel) and exhibit better overall fits (right panel), than the blue one with only exogenous terms. It is therefore very likely that the dyad-independent models have failed to capture the network dynamics key to the underlying data-generating process.

⁹For ERGM results AME estimates provide a better metric of effect sizes and significance than coefficient estimates, given the scaling problem affecting most incompletely specified models with omitted variables even uncorrelated to the main predictors. See [Duxbury \(2021\)](#) for an excellent exposition of the problem and its solutions.

¹⁰We note that the present results effectively decompose the original authors' results, which reports a pooled estimate for all female recipients of online mentions irrespective of the senders' gender(s). Their results show women to be more likely to receive mentions than men but not exactly from whom. This highlights the importance of using dyad-level covariates even within an already networked context, for more subtle and meaningful insights from inference.

¹¹As a simple illustration, for actors A, B, and C, reciprocal/mutual ties are of the form $A \leftrightarrow B$; transitive ties of the form $A \rightarrow B, A \rightarrow C, B \rightarrow C$; and for cyclic ties, $A \rightarrow B \rightarrow C \rightarrow A$.

Further results are shown in appendix section A.6.1. Among other things we note the exact equivalence in model estimates between the logistic and dyad-independent ERGM, as mentioned in section 2.3 above (table A.7); and the non-trivial differences in the way MPs interact with one another, with mutual exchanges being more likely than transitive ones and cyclic mentions unlikely to happen at all (figure A.5). These results in themselves give testimony to the methodological gains obtained from using a network-ERGM approach.

2.6 Case II: US Congressional Hearing Interruptions

The US data come from [Miller and Sutherland \(2022\)](#) and include directed interactions among 1,269 Congress members during committee hearings from the 105th to 115th Congresses (1994-2018). Each member is defined as being linked to another if she/he interrupted that person at least four times during congressional hearings.¹² For results legibility and over-time comparability concerns, the current exposition focuses on the 110th session in the middle of the period, although all analyses are extended to all eleven Congresses under coverage. For data and variables description see appendix sections A.2.2 and A.3.2, respectively.

Results are summarized in figure 2.2 and as before with AME plots shown on the left and ROC plots shown on the right. We see here that the two mixed-gender pairings both show marginally lower probabilities of having interruptions relative to the all-male baseline ($\widehat{AME}_{[M \rightarrow W] \vee [W \rightarrow M]} \approx [-0.5\%, -0.1\%]$), suggesting that women (men) are less likely to be interrupted by opposite- than same-gender colleagues during hearings, although such difference is only precisely estimated for the dyad-independent not dependent model; the difference between co-gender pairs is not significant in either case. These results stand in contrast to the original authors' with different ERGM specifications¹³ and suggest weak to no evidence for gender differences in interruption proneness. Similar to before, we also note that the inclusion of network-endogenous terms induces considerable improvement in model fit (right panel) whilst at the same time altering main effect estimates, albeit this time inflating both point and uncertainty estimates. The dyad-dependent models likely outperform the dyad-independent model in explaining target variations in the data as before.

¹²As discussed in the appendix, the results are insensitive to such threshold choices. In particular, directly modelling the raw count (exact number) of interruptions yields qualitatively similar results as when using a binarized measure as is currently done.

¹³The abovementioned scaling problem could be a main suspect, as the authors compute gender-pair effects as absolute level of, not change in, probability of interruption at given values of the main predictors, which is more prone to this issue. The AME is robust to the scaling problem and should be more reliable instead. Another more conservative conjecture is that the authors use an interaction model with both sender and receiver gender and gender pairing being interacted with other covariates, which could complicate results interpretation on their own and comparison with the present ones.

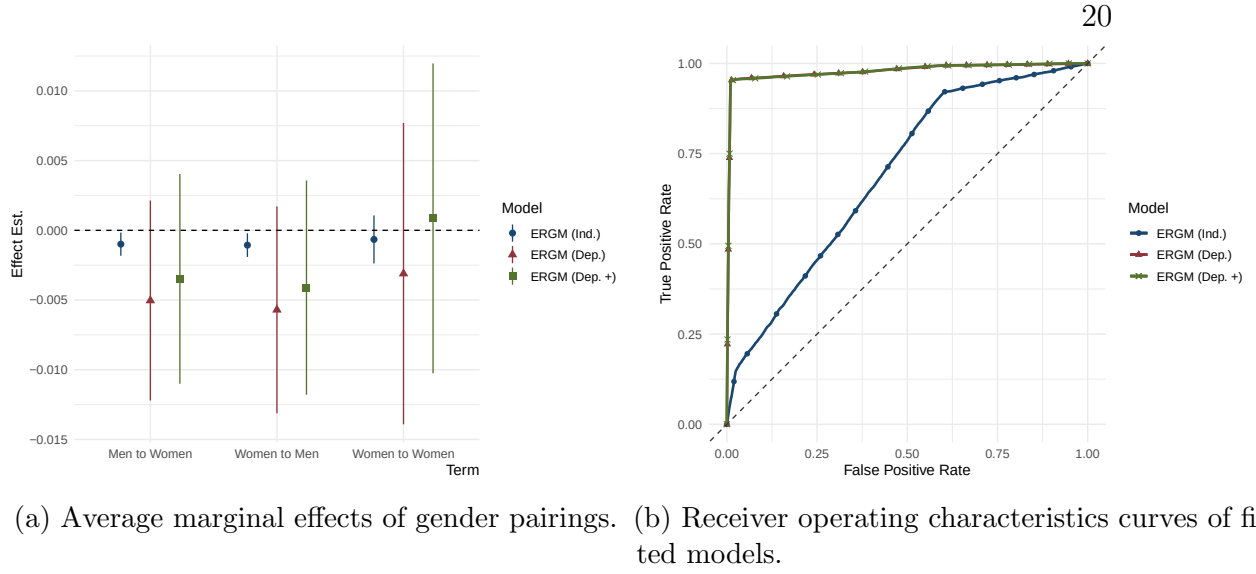


Figure 2.2: Summary of model results for the US Congressional hearing interruptions data.

Further results are shown in appendix A.6.2. Again we note two important patterns: one is the consistent equivalence of logistic and dyad-independent ERGM across all model estimates and their underperformance to the dyad-dependent ERGM (tables A.10, A.17, and A.14 to A.16), and the other is the strong role of reciprocity in driving members' interruption behavior against each other, typically around 20% in magnitude across all Congresses – indicating that being interrupted increases a member's willingness to strike back and interrupt the culprit by nearly 20% on average! These again make a strong case for the adoption of a network-ERGM approach in modelling deliberative process, on top of the Dutch case above.

2.7 Discussion

This note has introduced, explained, and demonstrated the methodological promise and utility of a network-ERGM approach to studying political deliberation. Its better measurement and modelling capacities make the approach worth advocating to applied researchers working on related topics. Two important limits exist for the proposed approach: one is its requirement for *edge*-level data, such as Twitter mentions and committee interruptions used here, without which the graphical structure cannot be established. The second is its demand on computational power: the simulation-based approach to solving $\kappa(\theta)$ (model 2.1 above) often requires large-scale sampling and lengthy runs that could be prohibitive to less powerful devices. It must be noted however that these are common issues in applied research, data availability and computational capacity respectively, and not unique to the proposed approach alone.

A further and less easily resolved issue concerns model dependency, namely, that results for the same relationship under testing could differ depending on model specification even with the exact same data and the same chosen model type. This is partly encountered in the two demonstrative cases above as well: in the Dutch study, the present dyad-level measure of gender (as pairs) yields more subtle and sender-dependent estimates of gender bias than the original authors' using a simple node-level measure; while in the US study the simplified linear additive models (up to a link function transformation) yields less impressive results about gender bias compared to the original authors' using more complex interaction specification. This is again not an issue unique to the present comparison between network and classical regression models as such, but affects nearly all observational studies to various extents; nonetheless, greater caution and better pre-modelling consideration are merited as a result. In this respect, relying on more informed theoretical analysis and engaging with a wider range of possible specifications, possibly with some selection and/or aggregation over these candidate models as well, may prove useful for mitigating the model dependency problem.

Looking forward, possible extensions to the approach could include, among many possible venues: (1) the use of more sophisticated network models, especially Bayesian nonparametric models, to even more flexibly and robustly model the data processes at play; (2) a more systematic definition and measurement of edges, in particular, what activity or aspect of the deliberative session counts as an edge and how best to represent it mathematically to allow for theoretically grounded empirical inference. For point (2), a promising approach might be pooling across multiple edge measures each defining its own subgraph.

Appendix A

Appendix to Chapter 2 (Paper I)

A.1 Exponential Random Graph Models

This section provides further discussion on three important issues in ERGM application: model estimation, goodness of fit, and limitation. A brief introduction on extending the model from binary to valued cases is also provided at the end. Note that as in the main text, the following graph terms are used exchangeably throughout the rest of the appendix: “graph” and “network”, “vertex” and “node”, and “edge” and “tie” (occasionally, “link” too).

As a reminder, a (binary) ERGM takes on the following graph- and tie- level forms, respectively:

$$P(X|\theta) = \frac{1}{\kappa(\theta)} \exp \{ \theta^T s(X) \}, \quad (\text{A.1})$$

and

$$P(X_{ij} = 1 | n_V, X_{ij}^C) = \text{logit}^{-1} \left\{ \sum_{k=1}^K \theta_k \sigma[s_k(x)] \right\}, \quad (\text{A.2})$$

where X is the adjacency matrix with $X_{ij} = 1$ if non-identical nodes i, j are connected and $X_{ij} = 0$ otherwise. n_V is the number of nodes in the network, X_{ij}^C all the dyads other than X_{ij} , $s(X)$ the vector of sufficient statistics with coefficients θ and change statistics $\sigma[.]$, and $\kappa(.)$ the normalizing constant ensuring the distribution integrates to unity. Further, denote $\mathcal{X}$ as the obtainable set of all possible networks with $X \in \mathcal{X}$.

A.1.1 Estimation

Since calculating the denominator in model (A.1) requires summing over *all* possible (discrete numbers of) networks, an analytical approach is generally unfeasible.¹ In practice, numerical approaches are usually taken to approximate the maximum likelihood estimate (MLE). Below presents a short summary of the general approach based on [Hunter et al. \(2008\)](#) and [Cranmer and Desmarais \(2011\)](#).

From (A.1) one can find the log-likelihood function as

$$\ell(\theta) = \theta^T s(X) - \log(\kappa(\theta, \mathcal{X})), \quad (\text{A.3})$$

using the same notations as in the generative model. It is preferred not to directly maximize this term but rather its derived log-ratio

$$\ell(\theta) - \ell(\theta_0) = (\theta - \theta_0)^T s(X) - \log\left(\frac{\kappa(\theta, \mathcal{X})}{\kappa(\theta_0, \mathcal{X})}\right), \quad (\text{A.4})$$

where θ_0 is some arbitrarily chosen parameters. Approximation of the difficult-to-obtain log-ratio term in (A.4) can be achieved by exploiting the following equality

$$\frac{\kappa(\theta, \mathcal{X})}{\kappa(\theta_0, \mathcal{X})} = \mathbb{E}_{\theta_0} [\exp\{(\theta - \theta_0)^T s(X)\}], \quad (\text{A.5})$$

with the expectation $\mathbb{E}_{\theta_0}$ taken on the assumption that $X \sim P_{\mathcal{X}, \theta_0}$, and using it in combination with the law of large numbers to reach the following approximating equation

$$\ell(\theta) - \ell(\theta_0) \approx (\theta - \theta_0)^T s(X) - \log\left[\frac{1}{M} \sum_{m=1}^M \exp\{(\theta - \theta_0)^T s(X_m)\}\right], \quad (\text{A.6})$$

where $X_1, \dots, X_M$ is a random sample taken from the distribution $\hat{P}_{\mathcal{X}, \theta_0}$ simulated using an MCMC sampling routine to be discussed shortly after. Theoretically with large enough M the choice of initial parameter values θ_0 should not matter; in realistic settings this turns out to be a non-trivial issue and may affect the precision of the approximation procedure. In particular, θ_0 should be sufficiently close to the true MLE ($\hat{\theta}^{\text{MLE}}$) lest the approximation via (A.6) fails. The default method for choosing the initial values used in the `ergm` package and adopted in the current analysis is pseudo-likelihood estimation, which yields a maximum pseudo-likelihood estimator (MPLE) through an alternative local approximation to the likelihood function. The

¹To see this, notice that for an undirected network with fixed n (dropping the vertex subscript $[v]$ for conciseness) the set of all obtainable works $\mathcal{X}$ may have up to $2^{\binom{n}{2}}$ possible networks, which blows up to $2^{\binom{n}{2} \times 2}$ if the network is directed. For even moderately sized n this can be a large quantity to estimate.

MPLE is easier to find than, and performs well as the initial values for, the true MLE. In fact in a dyad-independent ERGM, the MPLE and the MLE are the same.

The MCMC routine for finding θ follows the standard procedure from the MCMC approach. To briefly revisit, the sampling procedure seeks to construct a Markov chain on $\mathcal{X}$ with $\hat{P}_{\mathcal{X},\theta_0}(X = x)$ as the equilibrium distribution. Starting with a randomly drawn network in $\mathcal{X}$ a large number of networks are drawn following the appropriate Markov transition rule until (approximate) convergence to $\hat{P}_{\mathcal{X},\theta_0}(X = x)$ is achieved. Subsequent draws are then taken and used to form a (sequentially dependent) sample from the target model. Interested readers may seek further details about the MCMC approach from any standard textbooks on the topic. In the limit, as the number of transitions goes to infinity, convergence on the Markov chain is guaranteed under quite mild conditions, namely, *irreducibility* and *aperiodicity*. Somewhat oversimplified, the former refers to situations where all states can communicate with each other, and the latter for situations where returns to any state occur at irregular times.

Many types of chains for ERGM estimation are possible with different mixing properties. The default in `ergm` used here is the more general *Metropolis-Hastings algorithm*, which controls transition to a proposed network (x_{proposed}) via an auxiliary distribution dependent on the current network (x_{current}), such that the focus can be either on transitions or more spread out across $\mathcal{X}$. The probability of transition under the M-H algorithm is

$$\min \left\{ 1, \frac{P_{\mathcal{X},\theta_0}(X = x_{\text{proposed}})}{P_{\mathcal{X},\theta_0}(X = x_{\text{current}})} \frac{q(x_{\text{current}}, x_{\text{proposed}})}{q(x_{\text{proposed}}, x_{\text{current}})} \right\}, \quad (\text{A.7})$$

where $q(x_1, x_2)$ denotes the probability that $X_{\text{proposed}} = x_1$ given $X_{\text{current}} = x_2$ under the auxiliary distribution. As a side note, the *Metropolis algorithm* can be viewed as a special case of the general model (A.7) in which this auxiliary distribution is symmetric, i.e. $q(x_1, x_2) = q(x_2, x_1)$. The M-H algorithm is also open to modifications to reach easier and faster convergence.

Note that, beyond supplying the initial guesses for the MCMC-MLE approach the MPLE approach can also be used as an alternative method for parameter estimation. However, the MCMC-MLE approach is increasingly favored over the MPLE due to its proven abilities in making less biased and more efficient estimates with minor exceptions. All the models in this paper uses the MCMC approach as well. The chains have been run on the default settings of

the `ergm` package in R,² unless further fine-tuning is merited³ and yield the results reported throughout the main and current texts.

A.1.2 Goodness of Fit

One important issue with ERGM in practice is model degeneracy. Degeneracy is an estimation problem that affects badly specified models with poor fit, when the model lumps all or most of its probability mass on just one or a few possible graphs, typically extreme cases that are complete (all nodes fully connected) or empty (all unconnected) (Hunter et al. 2008; Cranmer and Desmarais 2011). Fully or nearly degenerate ERGM will lead to problems in MCMC estimation and yield unrealistic graphs like the ones just described.

A variety of tools exist to (*ex post*) diagnose potential degeneracy. A full discussion is beyond the scope of this appendix and readily available elsewhere, but a simple approach is to compare key network statistics from simulated networks based on the fitted model with the actual observed network. The underlying intuition relies on a known property of the exponential family, namely that a particular model where the $\hat{\theta}^{MLE}$ equals θ satisfies the following equality

$$\mathbb{E}_{\hat{\theta}^{MLE}}[s(X)] = s(x), \quad (\text{A.8})$$

i.e., the fitted model yields expectations for the network statistics that agree with their observed values. Notice that replacing $\hat{\theta}^{MLE}$ with θ in (A.8) leads to one of the two key assumptions of motivating ERGM (see main text section 2.3 for a quick reminder), that the observed values are the true expected values for these statistics. Using this method one may compare many relevant aspects of the network, such as the geodesic distance distribution (proportion of pairs of nodes with shortest connecting path equal to length $k = 1, \dots, K$), degree distribution, and so on. Tests against degeneracy of this type have been performed on all ERGM in this note using the default function in `ergm`, `gof`, with few strong signs detected, and due to the length of the outputs are not shown here. The test results are available from the author upon request.

²These include, among many other settings, a total of 1,024 MCMC steps (number of proposals) between sampled networks after 16,386 initial burn-ins, a maximum number of 60 times that parameter values are updated by maximizing sample-based MCMC likelihood, and a minimum MCMC sample size of 256 with a minimum of 32 for each parameter under estimation.

³For instance when parallel processing is needed for processing large networks or when the maximum number of iterations has to be reduced given time constraints. Thankfully, the goodness-of-fit statistics for these modified runs do not suggest major issues with model qualities that may invalid inference drawn therefrom.

A.1.3 Limitation

The ERGM certainly has its limitations as a modelling technique. The danger of model degeneracy as just introduced can be a major issue. However, as [Cranmer and Desmarais \(2011\)](#) have pointed out, degeneracy *per se* is not as severe a problem as it might appear, at least for empirically minded analysts: it can serve as a useful diagnosis for models that are poorly specified. That is, the fact that the model does not fit the data well is a natural indication — albeit only *post hoc* — that it is not sufficiently representative of the true process that have generated the data. In this sense, model degeneracy can help guard against previous misspecification and inform subsequent calibration.

Another main limit concerns computational demands. The MCMC-based estimation procedure often requires considerable computational power, which can be quite demanding for even moderately sized networks. The issue is more pertinent perhaps for researchers with more limited resources and/or under strict time constraints, especially graduate students with low setup personal devices and tight institutional deadlines. The issue is of course of a broader nature and affects most simulation-based inference techniques, Bayesian models in particular, and should not be taken as an inherent hindrance to the ERGM's application either. Access to high-performance cluster computers, use of cloud computing, optimizing and parallelizing running scripts are some common and increasingly popular workarounds in computationally intensive situations like this. Other issues like missing data can be discussed elsewhere and of less immediate relevance for the current study.

A.1.4 Extension to valued graphs

The ERGM has also been extended to valued networks where edge counts take on more than simple $\{0, 1\}$ values. The present section offers a cursory introduction to the basic case with linear mappings based on [Krivitsky \(2012\)](#). Redefine $\mathcal{X}$ as a set of mappings that assigns to each dyad (i, j) a count, such that $X_{ij} \in \mathbb{N}_0$ where $\mathbb{N}_0$ is the set of natural numbers plus zero. Then the simple⁴ graph-level model can be written as

$$P(X|\theta) = \frac{1}{\kappa(\theta)} h(X) \exp \{ \theta^T s(X) \}, \quad (\text{A.9})$$

where $h : \mathcal{X} \rightarrow [0, \infty)$ is the *reference measure* which sets the distribution relative to which the exponential form is specified. It has considerable importance for the valued model as it determines both the baseline shape of the dyad distribution and constraints on the parameter

⁴”Simple” refers to the stated fact that only linear valued ERGM are considered in this section. ”Curved” ERGM where the parameter space is mapped non-identically to some higher-dimensional space is not considered here.

space. For instance, when $h(x) = \prod_{ij}(x_{ij}!)^{-1}$, then for a one-parameter model its PMF (using (A.9)'s expression) becomes

$$P(X = x|\theta) = \frac{1}{\kappa(\theta)\prod_{ij}x_{ij}!}\exp\left\{\theta\sum_{ij}x_{ij}\right\} = \prod_{ij}\frac{\exp\{\theta x_{ij}\}}{x_{ij}!\exp\{\theta\}}, \quad (\text{A.10})$$

leading to $X_{ij} \stackrel{\text{i.i.d.}}{\sim} \text{Poisson}(\mu = \exp(\theta))$ with $\Theta = \mathbb{R}$. Model (A.10) is thus called the *Poisson-reference* ERGM. Analogous to the equivalence of the binary ERGM and logistic regression under dyadic independence before, the Poisson-reference ERGM reduces to a Poisson regression model when the sufficient statistics are linear in the focus dyad and do not depend on non-focus dyads, with dyad distribution given by

$$X_{ij} \stackrel{\text{ind.}}{\sim} \text{Poisson}\left(\mu = \exp\left\{\theta^T \Delta_{ij}^{0 \rightarrow 1} s(x)\right\}\right), \quad (\text{A.11})$$

where $\Delta_{ij}^{0 \rightarrow 1} s(x)$ is the vector of discrete change statistics (defined below) when the focus dyad has its edge count increased by 1. (When the latter are linear in the focus dyad but do depend on other dyads, then the model gives a dyad distribution that is only conditionally Poisson but not marginally so.)

The discrete change statistics generalizes the simple tie-toggling change statistics in (2.2) to the valued case to accommodate multiple possible levels of change in the edge count of a given dyad:

$$\Delta_{ij}^{k_1 \rightarrow k_2} s(x) \equiv s(x_{ij} = k_2) - s(x_{ij} = k_1), \quad (\text{A.12})$$

which represent the change in all sufficient statistics when one fixes all but the focus dyad to their observed values and allow for the focus dyad to change from k_1 to k_2 in its value. It can be shown that (A.12) results from taking the ratio of the probabilities of two networks that are identical except for the focus dyad's value ($k_2 - k_1$). The just-seen $\Delta_{ij}^{0 \rightarrow 1} s(x)$ is a special case when the difference is set to 1 for all statistics. For a Poisson-reference ERGM, the discrete change statistics can be used to assess the effect of a particular model term on the *decay rate* of the ratios of probabilities of successive values of the dyads and on the shape of their conditional distribution. Distinction between dyad-dependent and independent terms and issues like computational complexity and degeneracy are similar to those discussed in the binary case earlier, with necessary adjustments to the non-binary situation. Further extensions to the valued ERGM can also enable it to deal with issues like zero inflation (sparsity) and/or overdispersion. Like before, interested readers are invited to refer to the rapidly growing literature on this topic.

In the “Further results” section below, each of the main models for the two datasets have been re-run with non-binary edge values using the Poisson-reference model. This is both to check the robustness of the main results reported in main text section 2.4 and to make full use of the information provided by the data. The binary edge values in both datasets actually come from a “thinning” process where count values have been binarized under some arbitrary criteria. See subsequent section A.2 for details about data pre-processing. We notice little substantive change to the main insights yielded by binary models. An interesting additional insight only attainable from the valued models and worth early mentioning here is the fact that both deliberative processes, Dutch MPs’ Twitter interactions and US Congress Members’ committee interruptions, exhibit classical patterns of network sparsity: two randomly chosen actors are unlikely to interact with each other, but if they do then they will do so multiple times. This is revealed by the negative and statistically significant ($p < 0.001$) estimates for the zero-inflation term (“Nonzero”) in each case. As a conventional benchmark, results from companion zero-inflation models suggest similar patterns (tables A.8 and A.12).

A.2 Data Description

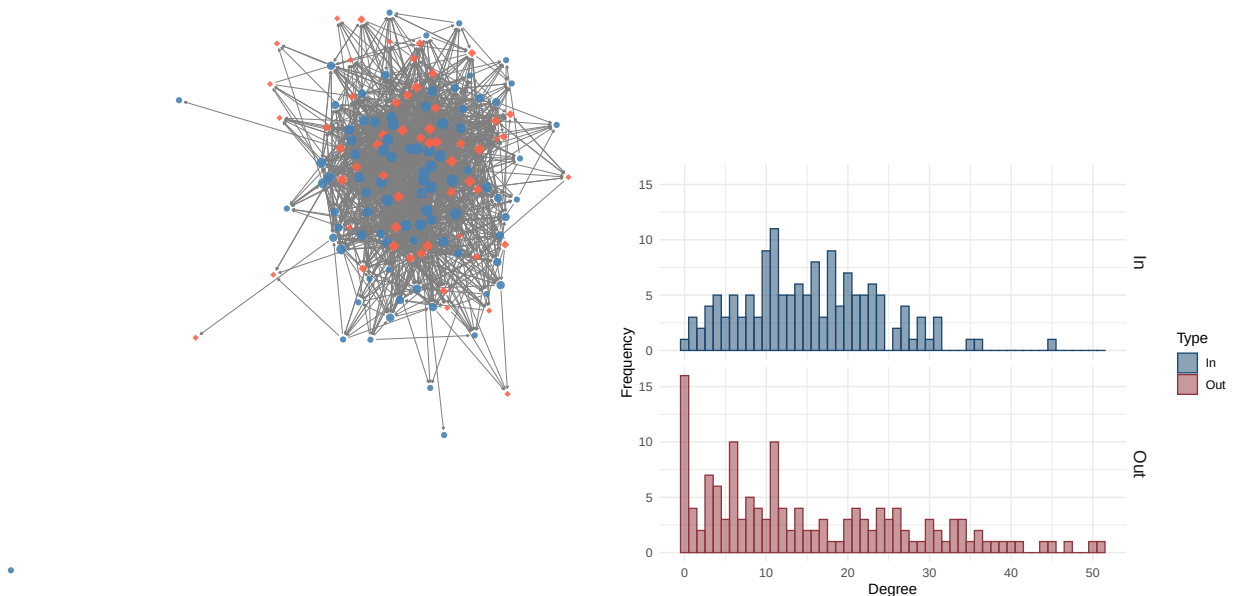
This section provides detailed information on the data used in the empirical analysis reported in main text sections 2.5 and 2.6.

A.2.1 The Dutch MP Twitters data

The Dutch data used in the first analysis (section 2.5) have been collected by [Del Valle, Broersma and Ponsioen \(2022\)](#).⁵ The authors collected a one-year sample of all tweets posted by 144 Dutch MPs (members of the House of Representatives [*Tweede Kamer der Staten-Generaal*]) using the social media software package *Coosto*. The sampling period was between 3 November 2015 and 3 November 2016. A total of 131,963 tweets were collected during the period. The authors then parsed the tweet texts through *Python* (version 3.7) to extract instances where one MP mentioned another. A total of 7,356 mentions were extracted from the tweets. An adjacency matrix was then created based on the extracted mentions and led to the network used in the current analysis. In this network the nodes are the MPs and the edges their mentions of each other.

Since the edges are counts of MP’s mentions, the network can be defined as either a binary or a valued graph. In the binary case, any MP who mentioned another MP ($X_{ij} > 0$ for two MPs i, j) are defined as sharing a tie and none otherwise ($X_{ij} = 0$). In the valued case, the

⁵Available online at https://osf.io/h63ba/?view_only=8f3ad2138fc6422f9508b797196dc6b5.



(a) Force-directed representation of the network (placed with Fruchterman-Reingold algorithm). Men in blue and women in red.

(b) Degree distribution in the network.

Figure A.1: Visualizations of the Dutch Twitter mentions network.

edge values are kept as they are, i.e., the total number of mentions by one MP of another. All analyses reported above and below using “binary” ERGM use the former type; all those using “valued” ERGM use the latter type.

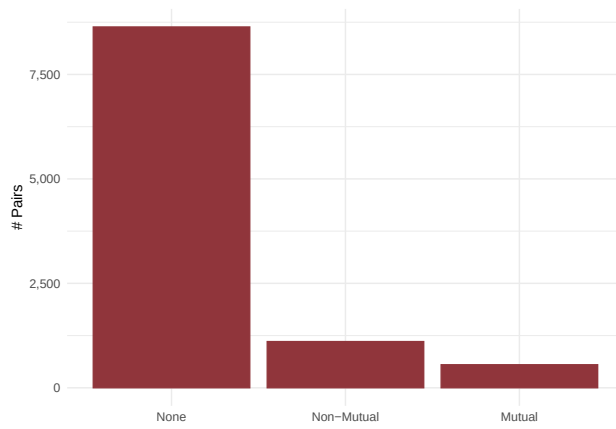
For a quick look at the (original, valued) network, figure A.1 shows the overall layout and degree distribution of the network, and figure A.2 shows the dyadic and triadic distribution in this network, respectively. For more information about the network, tables A.1 and A.2 present the key summary statistics at the network and vertex levels, respectively.

Table A.1: Key network statistics at the graph level.

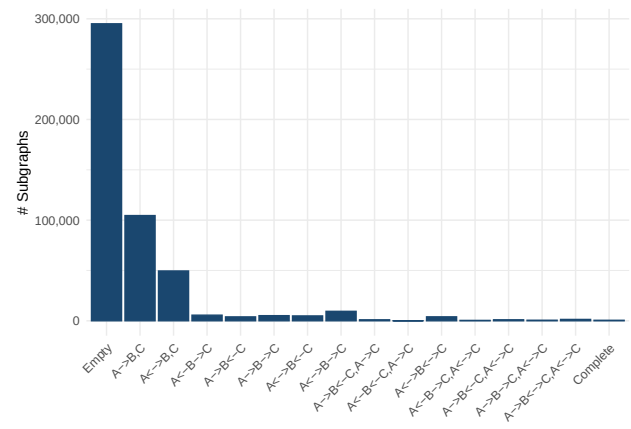
Statistic	Value
Order	144.00
Size	2211.00
Density	0.11
Reciprocity	0.50
Centrality	0.04
Path Length	2.65

Table A.2: Key network statistics at the vertex level.

Statistic	2.5%	25%	50%	75%	97.5%
Degree (In)	1.58	10	15.00	21.00	31.00
Degree (Out)	0.00	5	11.00	24.25	44.42
Closeness	0.00	0	0.00	0.00	0.00
Betweenness	0.00	0	0.01	0.01	0.03



(a) Distribution of dyads.



(b) Triad census results.

Figure A.2: Dyadic and triadic distributions in the Dutch Twitter mentions network.

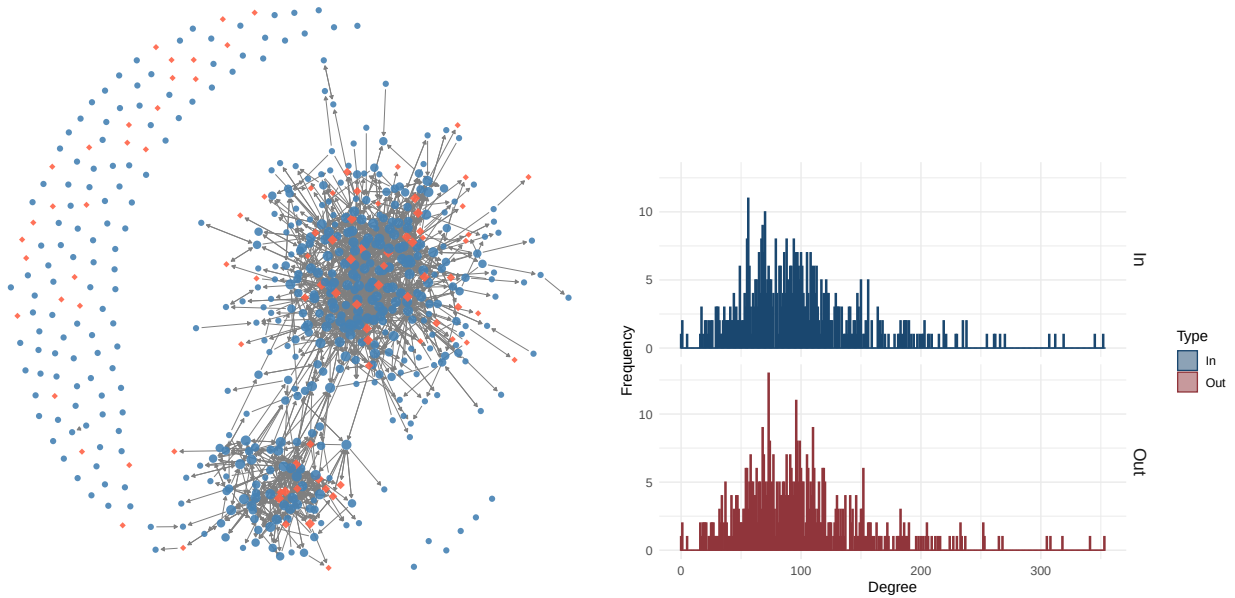
A.2.2 The US Congressional hearings data

The US data used in the second analysis (section 2.6) have been collected by [Miller and Sutherland \(2022\)](#).⁶ The authors analyzed 24,103 congressional hearings published by the Government Printing Office (GPO) between 1994 and 2018 (105th-115th Congresses), and merged this data with (a) an extended database of members' committee assignments (Stewart 2017) and (b) NOMINATE ideological data (Lewis et al. 2020). The resulting dataset include speech from 1,269 Members of Congress. The authors then used a series of regular expressions to process these hearing transcripts into annotated chunks of sequenced dialogue, and relied on the GPO's natural composition and annotation of the extracted chunks to extract interruptions, where an interruption is formally coded for in a chunk if the GPO signal was present within the last 10 characters of that chunk. Their approach produced a total of 3,081,247 chunks used in the author's own analysis, 152,338 of which ended in interruptions (4.9%). They then used the chunk-level data to produce networks for each of the eleven Congresses, where the nodes are the MPs and the edges their interruptions of each other. A total of 1,264 member nodes and 8,206,743 directed edges were prepared in this way. The authors further applied a "thinning rule" with a binarization threshold of four interruptions and ultimately obtained 174,508 nonzero edges for the eleven networks.

Note that despite the reported complete binarization, upon inspecting the data it is discovered that there are still multiple instances where two members shared more than one (supposedly thinned) edge. This is deemed valuable information that likely implies higher frequencies of pre-binarization interruptions between implicated members and used to construct the valued counterpart to the default binary network used in the authors' and current main analyses. In previous and subsequent sections all results using "binary" ERGM come from the default binary type; all those using "valued" ERGM come from the reconstructed valued type. As explained in the main text, the middling 110th Congress is chosen as the case under spotlight for main analysis, although all analyses have been replicated on all other Congresses as well.

Similar to before, for a quick view of the (default, binary) networks, figure A.3 shows the overall layout and degree distribution of the 110th network, and figure A.4 shows the degree distribution for all networks, respectively. For more information about the focus (110th) network, tables A.3 and A.4 present the key summary statistics at the network and vertex levels, respectively.

⁶Available online at: <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/5PEQSS>.



(a) Force-directed representation of the network (placed with Fruchterman-Reingold algorithm). Men in blue and women in red.

(b) Degree distribution in the network.

Figure A.3: Visualizations of the US Congressional hearings network.

Table A.3: Key network statistics at the graph level.

Statistic	Value
Order	547.00
Size	53951.00
Density	0.18
Reciprocity	0.95
Centrality	0.02
Path Length	0.00

Table A.4: Key network statistics at the vertex level.

Statistic	2.5%	25%	50%	75%	97.5%
Degree (In)	22.65	65	91	121	231.75
Degree (Out)	21.00	65	92	119	231.05
Closeness	0.00	0	0	0	0.00
Betweenness	0.00	0	0	0	0.02

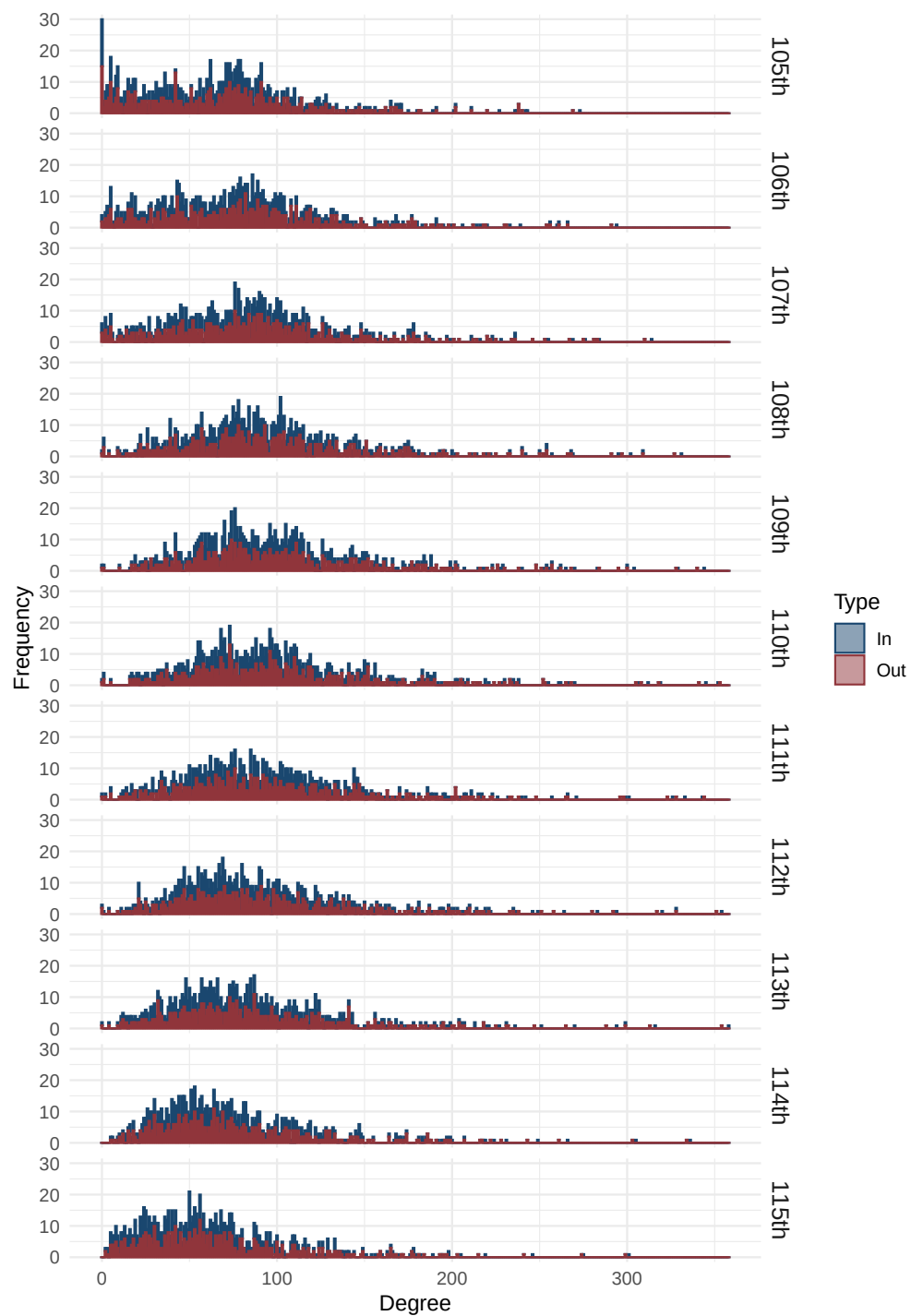


Figure A.4: Degree distribution in all Congressional hearings networks.

A.3 Variable Description

This section provides a description of the variables used in the empirical analysis reported in main text sections 2.5 and 2.6.

A.3.1 Dutch Twitter data

The variables used in the Dutch case are shown in list A.3.1.

- Gender: gender of each MP. 1 = female and 0 = male.
- Age: age in years of each MP.
- Party: partisan affiliation of each MP.
- Parliamentary commissions: total number of parliamentary commissions each MP has participated in.
- Seniority: total number of days each MP has spent in the chamber.
- Geographical area: residential area of each MP. 1 = residing in the *Randstad* (a megalopolis consisting of the four largest cities, Amsterdam, Rotterdam, the Hague, and Utrecht) and 0 = otherwise.

A.3.2 US Congress data

The variables used in the US case are shown in list A.3.2.

- Gender: gender of each Member of Congress. 1 = female and 0 = male.
- Party: partisan affiliation of each member. 1 = Republican and 0 = Democrat.
- Chamber: chamber membership of each member. 1 = the Senate and 0 = the House of Representatives.
- State: state represented by each member.
- Seniority: total number of years each Member has served in Congress.
- Ideology: NOMINATE score (first dimension) of each member in each Congress. Larger (more positive) values = more right-leaning and smaller (more negative) values = more left-leaning.

A.4 Summary Statistics

This section presents summary statistics for the variables used in the empirical analysis reported in main text sections 2.5 and 2.6 and introduced in appendix sections A.3.1 and A.3.2 above.

A.4.1 Dutch Twitter data

The summary statistics for the Dutch data are presented in table A.5 below.

Table A.5: Summary statistics of main variables.

Statistic	N	Mean	St. Dev.	Min	Max
Age	144	46.771	8.287	28	66
Female	144	0.396	0.491	0	1
Parl. Commissions	144	2.250	1.580	0	9
Seniority	144	2,243.451	1,340.099	9	6,743
Geo. Area (Metro)	143	0.294	0.457	0	1

Note: Variable “party” omitted from the table due to too many categories.

Table A.6: Summary statistics of main variables, all eleven Congresses (105th to 115th).

Congress	% Women		% Republican		% Senate		Seniority		Ideology	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
105	0.118	0.323	0.521	0.500	0.222	0.416	6.069	5.972	0.033	0.407
106	0.122	0.328	0.518	0.500	0.234	0.423	6.475	6.060	0.031	0.407
107	0.136	0.343	0.511	0.500	0.228	0.420	6.771	6.194	0.030	0.409
108	0.139	0.346	0.523	0.500	0.225	0.418	6.961	6.274	0.040	0.409
109	0.152	0.359	0.539	0.499	0.217	0.413	7.358	6.452	0.051	0.416
110	0.166	0.373	0.466	0.499	0.218	0.413	7.353	6.928	0.011	0.416
111	0.171	0.377	0.411	0.492	0.203	0.402	7.137	6.686	-0.016	0.414
112	0.172	0.378	0.530	0.500	0.211	0.408	6.670	6.475	0.072	0.442
113	0.187	0.390	0.514	0.500	0.213	0.410	6.240	6.153	0.073	0.449
114	0.195	0.396	0.565	0.496	0.197	0.398	6.166	5.700	0.105	0.452
115	0.193	0.395	0.550	0.498	0.193	0.395	6.495	5.798	0.095	0.453

Note: Variable “state” omitted from the table due to invariance (all equals 50).

A.4.2 US Congress data

The summary statistics for the US data are presented in table A.6 below. The statistics are shown separately for each Congress under coverage.

A.5 Model Terms in ERGM

This section provides further details for the model terms given in generic forms in main text model (2.4). Note that in the original model all terms are defined in terms of their change statistics (using the δ -notation); to shift focus to measurement issues alone, they will be referred to in their general form (using the s -notation) instead in the following discussion whenever needed.

A.5.1 Dyad-independent terms

Dyad-independent terms are denoted by δ_{exo} in (2.4), in change statistics form. Recall that these are the terms that imply no dependence between dyads, such that the presence or absence of a tie does *not* dependent on the state of other ties in the network. These are typically some measure of nodal attributes and can be direct (for each node) or interactive (for each dyad). For all the analyses reported throughout this paper the interactive type is chosen to best capture the relational nature of the analyzed process, where the effect of a certain nodal attribute on the state of a directed edge depends both on its realization for the source (sender) and the target (receiver) node. For instance, it is more likely that a man has different degrees of willingness to interrupt a women than a fellow man, so the effect of gender on the occurrence of interruption depends on not one but both sides involved in this case. Interactive measures are usually constructed using simple mathematical operations to combine nodal attributes for both nodes in a dyad.

Let i, j ($i \neq j$) be two nodes in the network. Then for the Dutch data, the following exogenous terms are constructed from the original variables listed in (A.3.1) and used in its analysis:

$$\text{GenderPair}_{i,j} := \text{Gender}_i \times \text{Gender}_j \quad (\text{Gender}_k \in \{\text{Men}, \text{Women}\} \forall k), \quad (\text{A.13})$$

$$\text{PartyPair}_{i,j} := \text{Party}_i \times \text{Party}_j \quad (\text{Party}_k \in \{\text{PvdA}, \text{VVD}, \text{Others}\} \forall k), \quad (\text{A.14})$$

$$\Delta \text{Age}_{i,j} := \text{Age}_j - \text{Age}_i \quad (j \text{ is the sender and } i \text{ is the receiver}), \quad (\text{A.15})$$

$$\Delta \text{Seniority}_{i,j} := \text{Seniority}_j - \text{Seniority}_i \quad (j \text{ is the sender and } i \text{ is the receiver}), \quad (\text{A.16})$$

$$\Delta \text{ParlCommissions}_{i,j} := \text{ParlCommissions}_j - \text{ParlCommissions}_i \quad (j \text{ is the sender and } i \text{ is the receiver}), \quad (\text{A.17})$$

$$\text{GeoAreaPair}_{i,j} := \text{GeoArea}_i \times \text{GeoArea}_j \quad (\text{GeoArea}_k \in \{\text{Metro}, \text{Non-metro}\} \forall k). \quad (\text{A.18})$$

And for the US data, the following list of exogenous terms from the original variables listed in (A.3.2):

$$\text{GenderPair}_{i,j} := \text{Gender}_i \times \text{Gender}_j \quad (\text{Gender}_k \in \{\text{Men}, \text{Women}\} \forall k), \quad (\text{A.19})$$

$$\text{PartyPair}_{i,j} := \text{Party}_i \times \text{Party}_j \quad (\text{Party}_k \in \{\text{Democrat}, \text{Republican}\} \forall k), \quad (\text{A.20})$$

$$\text{ChamberPair}_{i,j} := \text{Chamber}_i \times \text{Chamber}_j \quad (\text{Chamber}_k \in \{\text{Senate}, \text{House}\} \forall k), \quad (\text{A.21})$$

$$\text{StateMatch}_{i,j} := \begin{cases} 1, & \text{if } \text{State}_i = \text{State}_j, \\ 0, & \text{otherwise} \end{cases} \quad (\text{State}_k \text{ is one of the 50 US states } \forall k), \quad (\text{A.22})$$

$$\Delta \text{Seniority}_{i,j} := \text{Seniority}_j - \text{Seniority}_i \quad (j \text{ is the sender and } i \text{ is the receiver}), \quad (\text{A.23})$$

$$\Delta \text{Ideology}_{i,j} := \text{Ideology}_j - \text{Ideology}_i \quad (j \text{ is the sender and } i \text{ is the receiver}). \quad (\text{A.24})$$

In addition, two basic terms are also constructed for the entire network. These are the edge count term (θ_E in (2.4)) for both binary and valued ERGM and a nonzero term used in the valued ERGM only. The nonzero term is defined as the number of edges (non-zero values) in the network:

$$s_{x \neq 0}(x) := \sum_{ij} \mathbb{1}_{x_{ij}} > 0, \quad (\text{A.25})$$

and is used to account for overdispersion (zero-side) discussed in section A.1.4 before. Note that a *negative* coefficient on this term suggests the presence of zero inflation in the data (i.e., “less non-zero edges [or, more zero edges] than expected”), contrary to the interpretation of the similar term in zero-inflated Poisson models (see section A.6 below) where a positive coefficient implies the same symptom.

A.5.2 Dyad-dependent terms

Dyad-dependent terms are denoted by δ_{endo} in (2.4), in change statistics form. Recall that these are the terms that imply dependence between dyads, such that the presence or absence of a tie does dependent on the state of other ties in the network. These terms are the defining features of a “true” network ERGM and capture the endogenous aspect of the data generating process. Examples include the number of mutual ties, the numbers of different types of triads, and many more. Due to time and computational power constraints, the present analysis chooses simple, non-curved terms for empirical demonstration. Interested readers are welcome to extend the reported models to include further, more complex network statistics to reach even more finegrained insights about the deliberative process under study.

For the Dutch data, its smaller size allows for the estimation of more complex endogenous terms. Three terms are included to capture the reciprocal and triad-closure dynamics in the data: mutuality and transitive and cyclic ties, respectively. For the binary ERGM, they take the forms of, in the order just mentioned:

$$s_{i \leftrightarrow j}^b(x) := \sum_{ij} x_{ij} x_{ji}, \quad (\text{A.26})$$

$$s_{i \rightarrow j, i \rightarrow k, j \rightarrow k}^b(x) := \sum_{ijk} x_{ij} x_{ik} x_{jk}, \quad (\text{A.27})$$

$$s_{i \rightarrow j \rightarrow k \rightarrow i}^b(x) := \sum_{ijk} x_{ij} x_{jk} x_{ki}. \quad (\text{A.28})$$

For the valued ERGM, various generalizations of the statistics in (A.26) have been proposed. This paper uses a relatively simple set of such schemes that are easier to interpret and implement, given by (A.29) below.

$$s_{i \leftrightarrow j}^v(x) := \sum_{ij} \min(x_{ij} x_{ji}), \quad (\text{A.29})$$

$$s_{i \rightarrow j, i \rightarrow k, j \rightarrow k}^v(x) := \sum_{jk} \min(x_{jk}, \max_i(\min(x_{ij}, x_{ik}))), \quad (\text{A.30})$$

$$s_{i \rightarrow j \rightarrow k \rightarrow i}^v(x) := \sum_{ij} \min(x_{ij}, \max_k(\min(x_{ik}, x_{kj}))). \quad (\text{A.31})$$

For the US data, due to its larger size and the consequent model complexity, only the mutuality ($s_{i \leftrightarrow j}$) term has been included in both binary and valued ERGM. Only the former can admit the two triadic terms of transitivity and cyclicity.

A.6 Further Results

This section presents results from further analysis conducted using the two datasets introduced above. They serve as complements and extensions to the main results seen in main text sections 2.5 and 2.6. For the Dutch data, full model results are displayed for the sole dataset; and for the US data, full model results are displayed for both the focus (110th) and the full set of Congresses (105th-115th), respectively. The two subsections A.6.1 and A.6.2 below will present results for each case in turn.

To facilitate comparison with conventional non-network models, binary logistic and Poisson regression models have also been fitted as reference points for the binary and valued ERGM, respectively. For the logit case, two types of models were fitted, an unweighted one where the binarized dyadic link data were used as the data matrix; and a weighted one where the edge counts (see section A.2 again for reminder) were used as regression weights to adjust upon the binarized link set. For the Poisson case, zero-inflated models were used to better account for the sparse nature of the observed link data.

More formally, the (binary) logistic regressions take the familiar form of⁷

$$P(y = 1|x) = \text{logit}^{-1}(x^T \beta), \quad (\text{A.32})$$

where $\text{logit}^{-1}(x^T \beta) = \frac{\exp(x^T \beta)}{1 + \exp(x^T \beta)}$ is the inverse logit (logistic) link. x is the vector of exogenous predictors and β the coefficient vector. As explained in main text section 2.3, the ERGM reduces to logit models of form (A.32) whenever dyad-dependent terms are not involved on the right hand side. In the weighted case, the regression is further adjusted with a vector of weights w equal to the edge counts of the valued network under analysis (zero counts are set to one for nonnegativity, i.e., $w = \max(x_{ij}, 1)$).

The zero-inflated poisson regression is more complicated and involves a mixture-distribution situation. Following Zeileis, Kleiber and Jackman (2008), the zero-inflated density can be specified as a mixture of a point mass at zero $I_{\{0\}}(y)$ and a count distribution $f_{\text{count}}(y; x, \beta)$. The probability of observing a zero count is inflated with probability $\pi = f_{\text{zero}}(0; z, \gamma)$:

$$f_{\text{zeroinfl}}(y; x, \beta, \gamma) = f_{\text{zero}}(0; z, \gamma) \cdot I_{\{0\}}(y) + (1 - f_{\text{zero}}(0; z, \gamma)) \cdot f_{\text{count}}(y; x, \beta), \quad (\text{A.33})$$

where $I(\cdot)$ is the indicator function and the unobserved probability π of belonging to the point mass component is modeled by a binomial GLM $\pi = g^{-1}(z^T \gamma)$ with the logit link being the default link function. The corresponding regression equation for the mean is then

$$\mu_i = \pi_i \cdot 0 + (1 - \pi_i) \cdot \exp(x_i^T \beta), \quad (\text{A.34})$$

using the canonical log link. The vector of regressors in the zero-inflation model z_i and the regressors in the count component x_i need not to be distinct, with $z_i = 1$ is just an intercept

⁷Note that from here onward all model notations follow the conventional "y on the left, x on the right" rule (i.e., with y-letters indexing for the response and x-letters indexing for the predictors). In earlier sections, especially in the main text, where only the ERGM was considered, we have intentionally used x-notations on both sides of the model equation to stress the simultaneous nature of this type of models; for better clarity, consistency, and comparability with non-ERGM models in the present section we will switch back to the $y \sim x$ notation.

in the simplest case. In the current application z_i is indeed kept in this simple form to stay close to the valued ERGM where a corresponding “zero count” term is used that is similar in spirit. Estimation of model parameters are achieved by MLE. In terms of implementation, all zero-inflated Poisson models reported below have been estimated using the `zeroinfl` function from the `pscl` package in R on default settings.

Finally, for both logit and Poisson models, the same set of exogenous variables have been used between them and their corresponding ERGM counterparts to facilitate comparability. The dyad sets are the same as in the ERGM too. Network-specific measures regardless of dyadic (in)dependency are obviously unique to the latter only.

A.6.1 Dutch Twitter data

Table A.7 shows results for the logistic and three binary ERGM models. Figure A.5 plots the AME for four different types of ties — reciprocal, transitive, cyclical, and the baseline one-directional — to form between MPs. Table A.8 compares model results between zero-inflated Poisson and valued ERGM. Finally, table A.9 compares model results between unweighted and weighted logistic regressions.

For the binary ERGM, the model is specified as (A.35):

$$\begin{aligned}
 \text{logit} \{P(Y_{ij} = 1 | n_V, Y_{ij}^C)\} = & \sum_{d_{gender}=1}^{D_{gender}} \theta_{d_{gender}} [\text{GenderPair}]_{d_{gender_{ij}}} + \sum_{d_{party}=1}^{D_{party}} \theta_{d_{party}} [\text{PartyPair}]_{d_{party_{ij}}} \\
 & + \theta_{age} \Delta \text{Age}_{ij} + \theta_{senior} \Delta \text{Seniority}_{ij} \\
 & + \theta_{Parl} \Delta [\text{ParlCommissions}]_{ij} + \sum_{d_{geo}=1}^{D_{geo}} \theta_{d_{geo}} [\text{GeoAreaPair}]_{d_{geo_{ij}}} \\
 & + \theta_E + \mathbb{1}_{m_X \vee m_{XX}} \theta_{i \leftrightarrow j} \sigma_{i \leftrightarrow j}(y) \\
 & + \mathbb{1}_{m_{XX}} \theta_{i \rightarrow j, i \rightarrow k, j \rightarrow k} \sigma_{i \rightarrow j, i \rightarrow k, j \rightarrow k}(y) + \mathbb{1}_{m_{XX}} \theta_{i \rightarrow j \rightarrow k \rightarrow i} \sigma_{i \rightarrow j \rightarrow k \rightarrow i}(y),
 \end{aligned} \tag{A.35}$$

where m_X and m_{XX} refer to two types of dyad-dependent ERGM, the first (“ERGM X” in the tables below) with only reciprocity effects accounted for and the second (“ERGM XX” below) with both reciprocity and triad-closure (transitive and cyclic tie formation) effects accounted for.

For the valued ERGM, the model is specified as $Y_{ij} = y_{ij} | Y_{ij} \in \mathcal{Y}_{ij}(y) \stackrel{\text{ind.}}{\sim} \text{Poisson}(\mu_{ij})$ following the Poisson reference measure, where the mean part has the full form as (A.36):

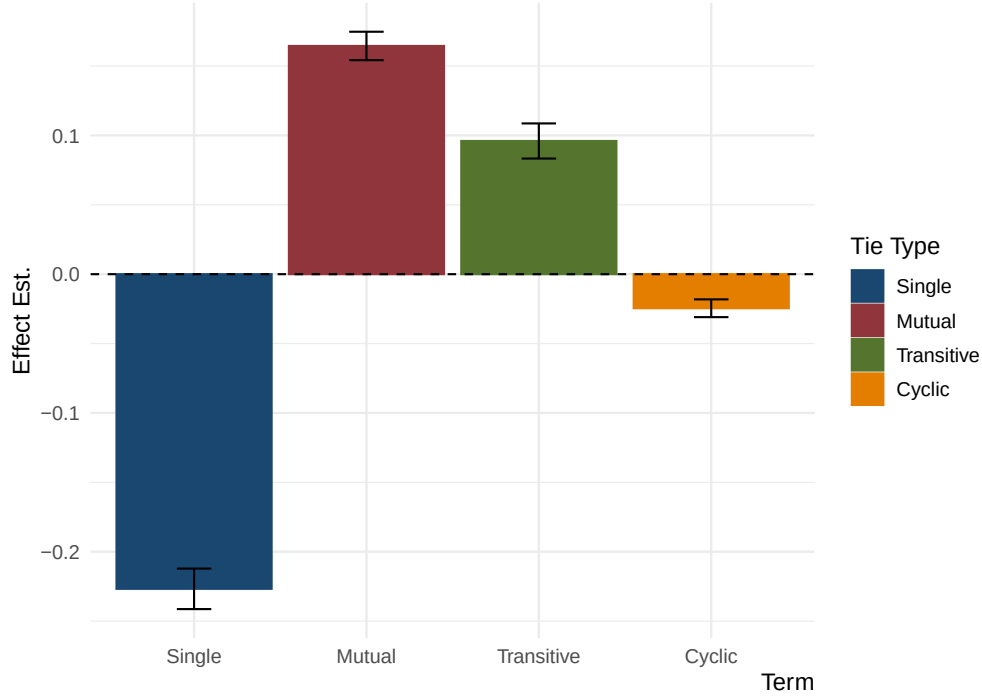


Figure A.5: Average marginal effects of tie types for Dutch Twitter data.

$$\begin{aligned}
\log \{\mu_{ij}\} = & \sum_{d_{gender}=1}^{D_{gender}} \theta_{d_{gender}} [\text{GenderPair}]_{d_{gender}ij} + \sum_{d_{party}=1}^{D_{party}} \theta_{d_{party}} [\text{PartyPair}]_{d_{party}ij} \\
& + \theta_{age} \Delta \text{Age}_{ij} + \theta_{senior} \Delta \text{Seniority}_{ij} \\
& + \theta_{Parl} \Delta [\text{ParlCommissions}]_{ij} + \sum_{d_{geo}=1}^{D_{geo}} \theta_{d_{geo}} [\text{GeoAreaPair}]_{d_{geo}ij} \\
& + \theta_E + \theta_{y \neq 0} s_{y \neq 0}(y) + \mathbb{1}_{m_X \vee m_{XX}} \theta_{i \leftrightarrow j} \sigma_{i \leftrightarrow j}(y) \\
& + \mathbb{1}_{m_{XX}} \theta_{i \rightarrow j, i \rightarrow k, j \rightarrow k} \sigma_{i \rightarrow j, i \rightarrow k, j \rightarrow k}(y) + \mathbb{1}_{m_{XX}} \theta_{i \rightarrow j \rightarrow k \rightarrow i} \sigma_{i \rightarrow j \rightarrow k \rightarrow i}(y), \quad (\text{A.36})
\end{aligned}$$

where m_X and m_{XX} refer to the two types of dyad-dependent ERGM like before, with reciprocity and reciprocity plus triad-closure terms each (“Valued ERGM X” and “Valued ERGM XX” below, respectively). As a reminder, the dyad-dependent terms are modified to accommodate the non-binary support as mentioned in section A.5.2.

The mean part of the logit and Poisson models are similar to their respective ERGM counterparts without the network-specific terms, so their full forms are omitted for conciseness.

Table A.7: Effect estimates of determinants of parliamentary Twitter mentions, logistic and ERGM results.

	Logistic	ERGM	ERGM X	ERGM XX
Intercept	−1.52*** (0.05)	−1.52*** (0.05)	−2.30*** (0.05)	−3.73*** (0.12)
Gender: Female to Male	−0.07 (0.06)	−0.07 (0.06)	0.01 (0.07)	−0.01 (0.06)
Gender: Male to Female	−0.19** (0.06)	−0.19** (0.06)	−0.20** (0.07)	−0.18** (0.06)
Gender: Female to Female	0.25*** (0.07)	0.25*** (0.07)	0.17** (0.06)	0.15** (0.05)
Party: Others to PvdA	−0.65*** (0.08)	−1.08*** (0.09)	−0.96*** (0.09)	−0.79*** (0.08)
Party: Others to VVD	−1.09*** (0.09)	−1.11*** (0.09)	−0.80*** (0.09)	−0.73*** (0.08)
Party: PvdA to Others	−1.08*** (0.09)	−0.65*** (0.08)	−0.26** (0.08)	−0.28*** (0.08)
Party: PvdA to PvdA	0.18* (0.09)	0.18* (0.09)	0.12 (0.07)	0.09 (0.07)
Party: PvdA to VVD	−1.70*** (0.15)	−1.20*** (0.12)	−0.70*** (0.13)	−0.68*** (0.12)
Party: VVD to Others	−1.11*** (0.09)	−1.09*** (0.09)	−0.76*** (0.09)	−0.66*** (0.08)
Party: VVD to PvdA	−1.20*** (0.12)	−1.70*** (0.15)	−1.41*** (0.16)	−1.11*** (0.13)
Party: VVD to VVD	0.22** (0.07)	0.22** (0.07)	0.15* (0.06)	0.09 (0.06)
Age: Difference	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Seniority: Difference	−0.00*** (0.00)	−0.00*** (0.00)	−0.00*** (0.00)	−0.00*** (0.00)
Parl. Commissions: Difference	0.02* (0.01)	0.02* (0.01)	0.04** (0.01)	0.04** (0.01)
Geo. Area: Metro to Non-metro	−0.16* (0.06)	−0.16* (0.06)	−0.12 (0.06)	−0.11 (0.06)
Geo. Area: Non-metro to Metro	−0.15* (0.06)	−0.15* (0.06)	−0.10 (0.06)	−0.09 (0.06)
Geo. Area: Metro to Metro	−0.01 (0.08)	−0.01 (0.08)	−0.02 (0.07)	0.01 (0.07)
Mutuality			2.64*** (0.08)	2.70*** (0.09)
Transitivity				1.58*** (0.11)
Cyclicity				−0.40*** (0.05)
Num. Obs	20592	20592	20592	20592
Log Likelihood	−6648.37	−6648.37	−6082.00	−5911.58
BIC	13475.53	13475.53	12352.72	12031.76

Table A.8: Comparing non-binary outcome models: zero-inflated Poisson vs valued ERGM fits.

	Zero-inflated Poisson	Valued ERGM	Valued ERGM X	Valued ERGM XX
Intercept	1.44*** (0.03)	1.29*** (0.02)	1.05*** (0.02)	0.99*** (0.02)
Gender: Female to Male	-0.01 (0.03)	-0.02 (0.02)	0.03* (0.02)	0.01 (0.02)
Gender: Male to Female	-0.27*** (0.04)	-0.10*** (0.02)	-0.12*** (0.02)	-0.09*** (0.02)
Gender: Female to Female	0.14*** (0.04)	0.09*** (0.02)	0.06*** (0.01)	0.06*** (0.01)
Party: Others to PvdA	-0.49*** (0.05)	-0.44*** (0.03)	-0.38*** (0.03)	-0.32*** (0.03)
Party: Others to VVD	-0.52*** (0.06)	-0.24*** (0.02)	-0.12*** (0.02)	-0.16*** (0.02)
Party: PvdA to Others	-0.88*** (0.07)	-0.26*** (0.02)	-0.11*** (0.03)	-0.16*** (0.02)
Party: PvdA to PvdA	-0.67*** (0.05)	-0.13*** (0.03)	-0.09*** (0.02)	-0.08*** (0.02)
Party: PvdA to VVD	-1.45*** (0.14)	-0.33*** (0.03)	-0.16*** (0.04)	-0.22*** (0.03)
Party: VVD to Others	0.07 (0.04)	-0.37*** (0.03)	-0.31*** (0.03)	-0.26*** (0.03)
Party: VVD to PvdA	-0.24** (0.07)	-0.65*** (0.06)	-0.58*** (0.06)	-0.48*** (0.06)
Party: VVD to VVD	-0.28*** (0.04)	-0.02 (0.02)	-0.01 (0.02)	-0.02 (0.02)
Age: Difference	-0.00 (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Seniority: Difference	-0.00*** (0.00)	-0.00*** (0.00)	-0.00*** (0.00)	-0.00*** (0.00)
Parl. Commissions: Difference	-0.01 (0.01)	0.00 (0.00)	0.01* (0.00)	0.01 (0.00)
Geo. Area: Metro to Non-metro	-0.13*** (0.04)	-0.06*** (0.02)	-0.05** (0.02)	-0.04* (0.02)
Geo. Area: Non-metro to Metro	-0.05 (0.03)	-0.05** (0.02)	-0.03 (0.02)	-0.04* (0.02)
Geo. Area: Metro to Metro	-0.31*** (0.05)	-0.07** (0.02)	-0.05* (0.02)	-0.06* (0.02)
Zero Inflation	1.99*** (0.02)			
Nonzero		-4.99*** (0.05)	-4.92*** (0.05)	-4.90*** (0.05)
Mutuality			0.88*** (0.03)	0.95*** (0.03)
Transitivity				0.24*** (0.02)
Cyclicalilty				-0.20*** (0.02)
Num. Obs	20592	20592	20592	20592
Log Likelihood	-14343.95	15339.42	15898.69	15850.16
BIC	28876.61	-30490.12	-31598.73	-31481.81

(1) *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

(2) Null model likelihood for valued ERGMS set to be zero. Likelihood-based inference only valid between these models.

Table A.9: Comparing logistic models: unweighted versus weighted fits.

	Unweighted	Weighted
Intercept	-1.52*** (0.05)	-0.08 (0.10)
Gender: Female to Male	-0.07 (0.06)	-0.08 (0.11)
Gender: Male to Female	-0.19** (0.06)	-0.40** (0.13)
Gender: Female to Female	0.25*** (0.07)	0.35** (0.12)
Party: Others to PvdA	-0.65*** (0.08)	-1.02*** (0.14)
Party: Others to VVD	-1.09*** (0.09)	-1.42*** (0.16)
Party: PvdA to Others	-1.08*** (0.09)	-1.67*** (0.14)
Party: PvdA to PvdA	0.18* (0.09)	-0.42** (0.13)
Party: PvdA to VVD	-1.70*** (0.15)	-2.33*** (0.21)
Party: VVD to Others	-1.11*** (0.09)	-1.00*** (0.17)
Party: VVD to PvdA	-1.20*** (0.12)	-1.32*** (0.22)
Party: VVD to VVD	0.22* (0.07)	-0.05 (0.15)
Age: Difference	0.00 (0.00)	0.00 (0.00)
Seniority: Difference	-0.00*** (0.00)	-0.00** (0.00)
Parl. Commissions: Difference	0.02* (0.01)	0.02 (0.02)
Geo. Area: Metro to Non-metro	-0.16* (0.06)	-0.25* (0.11)
Geo. Area: Non-metro to Metro	-0.15* (0.06)	-0.17 (0.13)
Geo. Area: Metro to Metro	-0.01 (0.08)	-0.30* (0.12)
Num. obs.	20592	20592
Log Likelihood	-6648.37	-13890.95
Pseudo R ²	0.05	0.08

(1) *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

(2) Edge weights used as regression weights in the weighted logistic fit.

A.6.2 US Congress data

Table A.10 presents the main model results using data from the 110th Congress. Table A.11 compares model results between binary and valued ERGM. Table A.12 compares model results between Poisson and valued ERGM. Note that due to extreme computational complexity, the valued ERGM could only admit the simpler reciprocity term, and no triadic term could be included in these models. Table A.13 compares model results between unweighted and weighted logistic regressions. These are all based on focus data from the 110th Congress.

For the entire set of Congress networks, tables A.14 to A.16 present results using binary ERGM with dyad-independent and dyad-dependent specifications, respectively. Specifically, table A.15 presents simpler results including only reciprocal tie effects, and table A.16 presents fuller results including both reciprocal and triadic tie effects. Figures A.6 and A.7 plot AMEs for gender pairings and tie types, respectively, for all eleven Congresses. Finally, tables A.17 and A.18 replicate the unweighted and weighted logit models in A.13 to the entire dataset, respectively, and table A.19 extends the zero-inflated Poisson model in A.12 to the entire dataset, too.

For the binary ERGM, the model is specified as (A.37):

$$\begin{aligned}
 \text{logit} \{P(Y_{ij} = 1 | n_V, Y_{ij}^C)\} = & \sum_{d_{gender}=1}^{D_{gender}} \theta_{d_{gender}} [\text{GenderPair}]_{d_{gender}ij} + \sum_{d_{party}=1}^{D_{party}} \theta_{d_{party}} [\text{PartyPair}]_{d_{party}ij} \\
 & + \sum_{d_{chamber}=1}^{D_{chamber}} \theta_{d_{chamber}} [\text{ChamberPair}]_{d_{chamber}ij} + \theta_{state} [\text{StateMatch}]_{ij} \\
 & + \theta_{senior} \Delta \text{Seniority}_{ij} + \theta_{ideo} \Delta \text{Ideology}_{ij} \\
 & + \theta_E + \mathbb{1}_{m_X \vee m_{XX}} \theta_{i \leftrightarrow j} \sigma_{i \leftrightarrow j}(y) \\
 & + \mathbb{1}_{m_{XX}} \theta_{i \rightarrow j, i \rightarrow k, j \rightarrow k} \sigma_{i \rightarrow j, i \rightarrow k, j \rightarrow k}(y) + \mathbb{1}_{m_{XX}} \theta_{i \rightarrow j \rightarrow k \rightarrow i} \sigma_{i \rightarrow j \rightarrow k \rightarrow i}(y),
 \end{aligned} \tag{A.37}$$

where m_X and m_{XX} refer to two types of dyad-dependent ERGM, the first (“ERGM X” or “dyad-dependent I” in the tables below) with only reciprocity effects accounted for and the second (“ERGM XX” or “dyad-dependent II” below) with both reciprocity and triad-closure (transitive and cyclic tie formation) effects accounted for.

For the valued ERGM, the model follows the same dyad-wise reduced form as in the Dutch case, except for the mean part now taking on a different set of predictors, as expressed in (A.38):

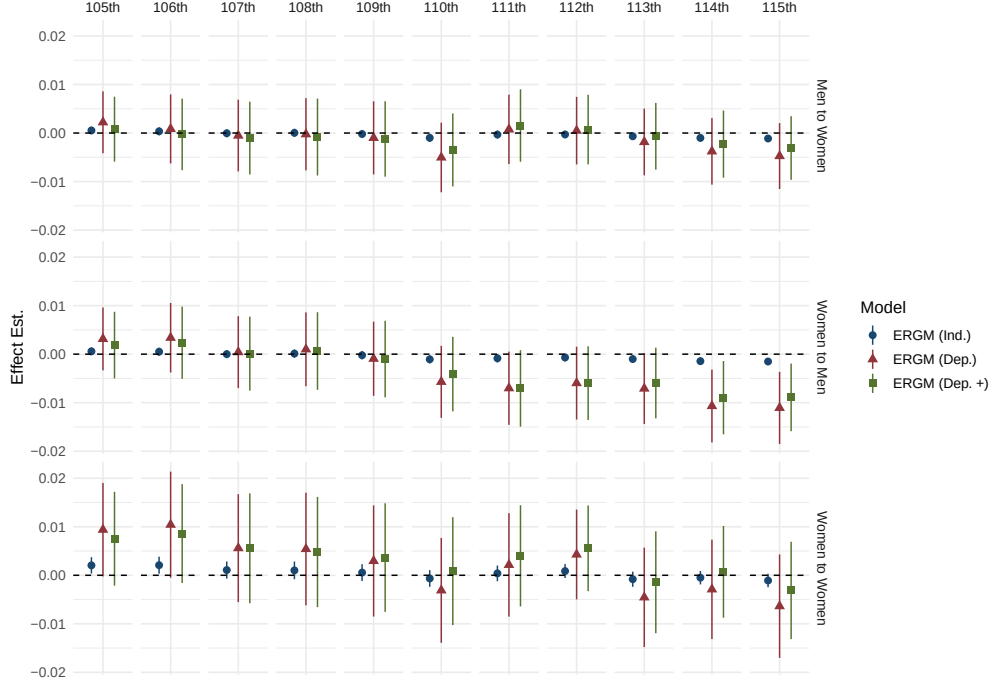


Figure A.6: Average marginal effects of gender pairings, all eleven Congresses.

$$\begin{aligned}
 \log \{\mu_{ij}\} = & \sum_{d_{gender}=1}^{D_{gender}} \theta_{d_{gender}} [\text{GenderPair}]_{d_{gender}ij} + \sum_{d_{party}=1}^{D_{party}} \theta_{d_{party}} [\text{PartyPair}]_{d_{party}ij} \\
 & + \sum_{d_{chamber}=1}^{D_{chamber}} \theta_{d_{chamber}} [\text{ChamberPair}]_{d_{chamber}ij} + \theta_{state} [\text{StateMatch}]_{ij} \\
 & + \theta_{senior} \Delta \text{Seniority}_{ij} + \theta_{ideo} \Delta \text{Ideology}_{ij} \\
 & + \theta_E + \theta_{y \neq 0} s_{y \neq 0}(y) + \mathbb{1}_{m_X} \theta_{i \leftrightarrow j} \sigma_{i \leftrightarrow j}(y),
 \end{aligned} \tag{A.38}$$

where m_X refers to the dyad-dependent ERGM (“Valued ERGM X” below) like before. The similarity of logit and Poisson models from the ERGM counterparts and their omission similarly apply here as before.

Table A.10: Effect estimates of determinants of congressional hearing interruptions, logistic and ERGM results.

	Logistic	ERGM	ERGM X	ERGM XX
Intercept	−5.29*** (0.06)	−5.29*** (0.06)	−5.80*** (0.05)	−6.02*** (0.06)
Gender: Female to Male	−0.19* (0.08)	−0.19* (0.08)	−0.13 (0.09)	−0.08 (0.08)
Gender: Male to Female	−0.18* (0.08)	−0.18* (0.08)	−0.12 (0.08)	−0.07 (0.08)
Gender: Female to Female	−0.12 (0.16)	−0.12 (0.16)	−0.07 (0.13)	0.02 (0.12)
Party: Rep. to Dem.	0.56*** (0.11)	0.56*** (0.12)	0.38* (0.15)	0.47** (0.15)
Party: Dem. to Rep.	0.53*** (0.12)	0.53*** (0.12)	0.36* (0.15)	0.46** (0.15)
Party: Rep. to Rep.	−0.02 (0.08)	−0.02 (0.08)	−0.01 (0.07)	−0.03 (0.06)
Chamber: Senate to House	−2.40*** (0.18)	−2.40*** (0.18)	−1.98*** (0.18)	−1.81*** (0.18)
Chamber: House to Senate	−2.37*** (0.19)	−2.37*** (0.19)	−1.84*** (0.19)	−1.65*** (0.18)
Chamber: Senate to Senate	1.49*** (0.06)	1.49*** (0.06)	0.95*** (0.05)	0.50*** (0.02)
State: Same	0.48*** (0.11)	0.48*** (0.11)	0.33*** (0.09)	0.30*** (0.08)
Seniority: Difference	0.01** (0.00)	0.01** (0.00)	0.02*** (0.00)	0.02*** (0.00)
Ideology: Difference	−0.01 (0.12)	−0.01 (0.13)	0.01 (0.18)	0.01 (0.17)
Mutuality			5.42*** (0.09)	4.47*** (0.11)
Transitivity				1.05*** (0.08)
Cyclicalilty				0.26** (0.08)
Num. Obs.	298662	298662	298662	298662
Log Likelihood	−9556.43	−9556.43	−166954.96	−132170.34
BIC	19276.74	19276.74	334086.43	264542.39

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.11: Comparing ERGMs: binary versus valued fits.

	Binary ERGM	Binary ERGM X	Binary ERGM XX	Valued ERGM	Valued ERGM X
Intercept	-5.29*** (0.06)	-5.80*** (0.05)	-6.02*** (0.06)	-2.00*** (0.09)	-2.14*** (0.06)
Gender: Female to Male	-0.19* (0.08)	-0.13 (0.09)	-0.08 (0.08)	-0.13* (0.07)	-0.06 (0.08)
Gender: Male to Female	-0.18* (0.08)	-0.12 (0.08)	-0.07 (0.08)	-0.14* (0.06)	-0.03 (0.07)
Gender: Female to Female	-0.12 (0.16)	-0.07 (0.13)	0.02 (0.12)	-0.07 (0.12)	-0.08* (0.03)
Party: Rep. to Dem.	0.56*** (0.12)	0.38* (0.15)	0.47** (0.15)	0.50*** (0.10)	0.33* (0.16)
Party: Dem. to Rep.	0.53*** (0.12)	0.36* (0.15)	0.46** (0.15)	0.48*** (0.10)	0.18 (0.16)
Party: Rep. to Rep.	-0.02 (0.08)	-0.01 (0.07)	-0.03 (0.06)	-0.04 (0.07)	-0.02 (0.03)
Chamber: Senate to House	-2.40*** (0.18)	-1.98*** (0.18)	-1.81*** (0.18)	-2.21*** (0.17)	-1.82*** (0.25)
Chamber: House to Senate	-2.37*** (0.19)	-1.84*** (0.19)	-1.65*** (0.18)	-2.15*** (0.17)	-1.56*** (0.30)
Chamber: Senate to Senate	1.49*** (0.06)	0.95*** (0.05)	0.50*** (0.02)	1.14*** (0.05)	0.84*** (0.02)
State: Same	0.48*** (0.11)	0.33*** (0.09)	0.30*** (0.08)	0.37*** (0.09)	0.20*** (0.03)
Seniority: Difference	0.01** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.01* (0.00)	0.01** (0.00)
Ideology: Difference	-0.01 (0.13)	0.01 (0.18)	0.01 (0.17)	0.01 (0.10)	-0.08 (0.21)
Mutuality		5.42*** (0.09)	4.47*** (0.11)		5.20*** (0.10)
Transitivity			1.05*** (0.08)		
Cyclicity			0.26** (0.08)		
Nonzero				-3.32*** (0.08)	-3.81*** (0.05)
Num. Obs.	298662	298662	298662	298662	298662

(1) *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

(2) Due to ongoing software issues, no log-likelihood based fit statistics are available for the valued ERGM at the moment.

Table A.12: Comparing non-binary outcome models: zero-inflated Poisson vs valued ERGM fits.

	Zero-inflated Poisson	Valued ERGM	Valued ERGM X
Intercept	−2.36*** (0.09)	−2.00*** (0.09)	−2.14*** (0.06)
Gender: Female to Male	−0.16* (0.08)	−0.13* (0.07)	−0.06 (0.08)
Gender: Male to Female	−0.16* (0.08)	−0.14* (0.06)	−0.03 (0.07)
Gender: Female to Female	−0.18 (0.16)	−0.07 (0.12)	−0.08* (0.03)
Party: Rep. to Dem.	0.60*** (0.12)	0.50*** (0.10)	0.33* (0.16)
Party: Dem. to Rep.	0.62*** (0.12)	0.48*** (0.10)	0.18 (0.16)
Party: Rep. to Rep.	−0.01 (0.08)	−0.04 (0.07)	−0.02 (0.03)
Chamber: Senate to House	−2.37*** (0.17)	−2.21*** (0.17)	−1.82*** (0.25)
Chamber: House to Senate	−2.30*** (0.17)	−2.15*** (0.17)	−1.56*** (0.30)
Chamber: Senate to Senate	1.42*** (0.06)	1.14*** (0.05)	0.84*** (0.02)
State: Same	0.45*** (0.11)	0.37*** (0.09)	0.20*** (0.03)
Seniority: Difference	0.01* (0.00)	0.01* (0.00)	0.01** (0.00)
Ideology: Difference	0.04 (0.13)	0.01 (0.10)	−0.08 (0.21)
Zero Inflation	2.80*** (0.07)		
Nonzero		−3.32*** (0.08)	−3.81*** (0.05)
Mutuality			5.20*** (0.10)
Num. Obs.	298662	298662	298662

(1) *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

(2) Due to ongoing software issues, no log-likelihood based fit statistics are available for the valued ERGM at the moment.

Table A.13: Comparing logistic models: unweighted versus weighted fits.

	Unweighted	Weighted
Intercept	−5.29*** (0.06)	−5.21*** (0.06)
Gender: Female to Male	−0.19* (0.08)	−0.17* (0.08)
Gender: Male to Female	−0.18* (0.08)	−0.17* (0.08)
Gender: Female to Female	−0.12 (0.16)	−0.09 (0.16)
Party: Rep. to Dem.	0.56*** (0.11)	0.61*** (0.12)
Party: Dem. to Rep.	0.53*** (0.12)	0.61*** (0.13)
Party: Rep. to Rep.	−0.02 (0.08)	−0.05 (0.09)
Chamber: Senate to House	−2.40*** (0.18)	−2.38*** (0.19)
Chamber: House to Senate	−2.37*** (0.19)	−2.30*** (0.20)
Chamber: Senate to Senate	1.49*** (0.06)	1.55*** (0.06)
State: Same	0.48*** (0.11)	0.47*** (0.11)
Seniority: Difference	0.01** (0.00)	0.01** (0.00)
Ideology: Difference	−0.01 (0.12)	0.03 (0.14)
Num. Obs.	298662	298662
Log Likelihood	−9556.43	−10604.71
Pseudo R ²	0.08	0.08

(1) *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

(2) Edge weights used as regression weights in the weighted logistic fit.

Table A.14: All ERGM fits (binary and dyad-independent): 105th to 115th Congresses.

	105th	106th	107th	108th	109th	110th	111th	112th	113th	114th	115th
Intercept	-5.68*** (0.08)	-5.51*** (0.07)	-5.50*** (0.07)	-5.30*** (0.07)	-5.21*** (0.07)	-5.29*** (0.06)	-5.49*** (0.06)	-5.33*** (0.07)	-5.48*** (0.08)	-5.74*** (0.10)	-5.97*** (0.11)
Gender: Female to Male	0.14 (0.10)	0.11 (0.09)	0.00 (0.09)	0.02 (0.08)	-0.04 (0.08)	-0.19* (0.08)	-0.16 (0.08)	-0.13 (0.08)	-0.21* (0.09)	-0.36*** (0.10)	-0.47*** (0.11)
Gender: Male to Female	0.13 (0.09)	0.08 (0.09)	-0.01 (0.09)	0.01 (0.08)	-0.04 (0.08)	-0.18* (0.08)	-0.06 (0.08)	-0.06 (0.08)	-0.14 (0.08)	-0.25** (0.09)	-0.35*** (0.11)
Gender: Female to Female	0.49* (0.21)	0.44* (0.19)	0.22 (0.18)	0.19 (0.17)	0.10 (0.16)	-0.12 (0.16)	0.07 (0.15)	0.17 (0.14)	-0.17 (0.17)	-0.12 (0.18)	-0.34 (0.22)
Party: Rep. to Dem.	0.69*** (0.14)	0.69*** (0.13)	0.75*** (0.13)	0.66*** (0.13)	0.54*** (0.13)	0.56*** (0.12)	0.63*** (0.12)	0.30* (0.14)	0.44** (0.14)	0.44* (0.17)	0.47* (0.19)
Party: Dem. to Rep.	0.43** (0.15)	0.39** (0.13)	0.49*** (0.13)	0.39** (0.13)	0.37** (0.13)	0.53*** (0.12)	0.81*** (0.12)	0.74*** (0.14)	0.77*** (0.14)	1.01*** (0.17)	1.03*** (0.19)
Party: Rep. to Rep.	-0.24* (0.10)	-0.19* (0.09)	-0.09 (0.09)	-0.26** (0.09)	-0.40*** (0.09)	-0.02 (0.08)	0.25** (0.09)	-0.04 (0.09)	0.27** (0.09)	0.48*** (0.11)	0.56*** (0.12)
Chamber: Senate to House	-2.19*** (0.20)	-2.33*** (0.19)	-2.47*** (0.20)	-2.26*** (0.17)	-2.31*** (0.18)	-2.40*** (0.18)	-2.22*** (0.18)	-2.20*** (0.17)	-2.00*** (0.16)	-1.86*** (0.17)	-1.72*** (0.18)
Chamber: House to Senate	-2.41*** (0.23)	-2.55*** (0.22)	-2.52*** (0.21)	-2.33*** (0.19)	-2.31*** (0.19)	-2.37*** (0.19)	-2.28*** (0.19)	-2.24*** (0.18)	-2.13*** (0.17)	-2.00*** (0.19)	-1.88*** (0.20)
Chamber: Senate to Senate	1.68*** (0.07)	1.49*** (0.06)	1.44*** (0.06)	1.47*** (0.06)	1.53*** (0.06)	1.49*** (0.06)	1.48*** (0.06)	1.16*** (0.07)	1.11*** (0.07)	0.72*** (0.10)	0.69*** (0.12)
State: Same	0.39** (0.13)	0.41*** (0.12)	0.45*** (0.12)	0.33** (0.12)	0.45*** (0.11)	0.48*** (0.11)	0.59*** (0.11)	0.62*** (0.11)	0.53*** (0.12)	0.57*** (0.13)	0.56*** (0.14)
Seniority: Difference	0.01* (0.00)	0.01* (0.00)	0.01* (0.00)	0.01** (0.00)	0.01** (0.00)	0.01** (0.00)	0.01* (0.00)	0.01* (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Ideology: Difference	-0.11 (0.16)	-0.15 (0.14)	-0.14 (0.14)	-0.16 (0.13)	-0.10 (0.14)	-0.01 (0.13)	0.12 (0.13)	0.31* (0.13)	0.26* (0.13)	0.40* (0.16)	0.38* (0.17)
Num. Obs	285690	281430	286760	283556	284622	298662	283556	291060	292140	279312	283556
Log Likelihood	-7143.35	-7777.08	-8219.11	-8914.49	-9080.55	-9556.43	-8871.77	-8842.53	-8378.76	-6965.51	-5930.34
BIC	14450.02	15717.28	16601.59	17992.19	18324.37	19276.74	17906.76	17848.61	16921.12	14094.05	12023.89

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.15: All ERGM fits (binary and dyad-dependent I): 105th to 115th Congresses.

	105th	106th	107th	108th	109th	110th	111th	112th	113th	114th	115th
Intercept	-6.18*** (0.07)	-6.03*** (0.07)	-6.00*** (0.07)	-5.82*** (0.06)	-5.73*** (0.06)	-5.80*** (0.05)	-5.96*** (0.06)	-5.93*** (0.07)	-6.08*** (0.07)	-6.35*** (0.09)	-6.60*** (0.10)
Mutuality	5.78*** (0.10)	5.69*** (0.10)	5.62*** (0.10)	5.46*** (0.09)	5.36*** (0.09)	5.42*** (0.09)	5.46*** (0.09)	5.75*** (0.09)	5.88*** (0.10)	6.15*** (0.11)	6.44*** (0.12)
Gender: Female to Male	0.10 (0.11)	0.10 (0.10)	0.01 (0.10)	0.02 (0.09)	-0.02 (0.09)	-0.13 (0.13)	-0.16 (0.09)	-0.14 (0.09)	-0.18 (0.10)	-0.30** (0.11)	-0.37** (0.13)
Gender: Male to Female	0.07 (0.11)	0.02 (0.10)	-0.01 (0.10)	-0.01 (0.09)	-0.02 (0.09)	-0.12 (0.08)	0.02 (0.09)	0.01 (0.09)	-0.05 (0.09)	-0.11 (0.10)	-0.16 (0.12)
Gender: Female to Female	0.31 (0.16)	0.29 (0.16)	0.15 (0.15)	0.13 (0.14)	0.07 (0.13)	-0.07 (0.13)	0.05 (0.13)	0.10 (0.11)	-0.12 (0.14)	-0.08 (0.15)	-0.21 (0.18)
Party: Rep. to Dem.	0.64*** (0.19)	0.66*** (0.17)	0.68*** (0.17)	0.62*** (0.16)	0.46** (0.16)	0.38* (0.15)	0.31 (0.16)	-0.11 (0.18)	0.06 (0.18)	-0.10 (0.22)	-0.09 (0.25)
Party: Dem. to Rep.	0.11 (0.19)	0.05 (0.17)	0.15 (0.17)	0.09 (0.16)	0.16 (0.16)	0.36* (0.15)	0.68*** (0.16)	0.79*** (0.18)	0.74*** (0.18)	1.07*** (0.22)	1.09*** (0.25)
Party: Rep. to Rep.	-0.17* (0.08)	-0.13 (0.08)	-0.06 (0.08)	-0.18* (0.07)	-0.28*** (0.07)	-0.01 (0.07)	0.18* (0.07)	-0.03 (0.07)	0.18* (0.07)	0.33*** (0.09)	0.38*** (0.10)
Chamber: Senate to House	-1.76*** (0.20)	-1.87*** (0.20)	-2.01*** (0.20)	-1.83*** (0.18)	-1.91*** (0.18)	-1.98*** (0.18)	-1.79*** (0.18)	-1.72*** (0.17)	-1.50*** (0.17)	-1.35*** (0.17)	-1.21*** (0.19)
Chamber: House to Senate	-1.89*** (0.23)	-2.02*** (0.22)	-1.97*** (0.21)	-1.80*** (0.19)	-1.79*** (0.19)	-1.84*** (0.19)	-1.77*** (0.19)	-1.68*** (0.18)	-1.61*** (0.18)	-1.49*** (0.20)	-1.40*** (0.20)
Chamber: Senate to Senate	1.06*** (0.05)	0.94*** (0.05)	0.91*** (0.05)	0.93*** (0.05)	0.98*** (0.05)	0.95*** (0.05)	0.95*** (0.05)	0.73*** (0.06)	0.70*** (0.06)	0.46*** (0.08)	0.43*** (0.09)
State: Same	0.26* (0.10)	0.28** (0.10)	0.30** (0.10)	0.22* (0.10)	0.30*** (0.09)	0.33*** (0.09)	0.39*** (0.09)	0.41*** (0.09)	0.35*** (0.09)	0.38*** (0.11)	0.36*** (0.11)
Seniority: Difference	0.02*** (0.01)	0.02*** (0.01)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.01* (0.00)	0.01 (0.01)	0.01 (0.01)
Ideology: Difference	-0.21 (0.22)	-0.29 (0.20)	-0.28 (0.20)	-0.30 (0.19)	-0.16 (0.19)	0.01 (0.18)	0.26 (0.18)	0.64*** (0.19)	0.55** (0.19)	0.85*** (0.23)	0.81** (0.26)
Num. Obs	285690	281430	286760	283556	284622	298662	283556	291060	292140	279312	283556
Log Likelihood	-117251.87	-132920.43	-146086.73	-157596.07	-165202.89	-166954.96	-157947.81	-152130.29	-141637.84	-127297.40	-112352.70
BIC	234679.62	266016.53	292349.39	315367.91	330581.60	334086.43	316071.40	304436.72	283451.86	254770.36	224881.17

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.16: All ERGM fits (binary and dyad-dependent II): 105th to 115th Congresses.

	105th	106th	107th	108th	109th	110th	111th	112th	113th	114th	115th
Intercept	-6.33*** (0.07)	-6.19*** (0.07)	-6.20*** (0.07)	-6.07*** (0.06)	-6.00*** (0.06)	-6.02*** (0.06)	-6.12*** (0.06)	-6.14*** (0.06)	-6.23*** (0.07)	-6.49*** (0.09)	-6.73*** (0.10)
Mutuality	5.00*** (0.12)	4.95*** (0.12)	4.84*** (0.11)	4.56*** (0.11)	4.46*** (0.11)	4.47*** (0.11)	4.53*** (0.11)	4.81*** (0.11)	5.15*** (0.11)	5.53*** (0.12)	5.91*** (0.14)
Transitivity	0.90*** (0.12)	1.00*** (0.11)	1.12*** (0.10)	1.07*** (0.08)	1.01*** (0.08)	1.05*** (0.08)	1.07*** (0.08)	1.05*** (0.09)	1.00*** (0.10)	1.20*** (0.13)	1.44*** (0.15)
Cyclicity	0.37*** (0.12)	0.20 (0.11)	0.10 (0.10)	0.22* (0.09)	0.28*** (0.08)	0.26** (0.08)	0.22* (0.09)	0.20* (0.09)	0.17 (0.11)	-0.04 (0.13)	-0.27 (0.15)
Gender: Female to Male	0.06 (0.10)	0.06 (0.10)	0.00 (0.09)	0.01 (0.09)	-0.02 (0.08)	-0.08 (0.08)	-0.15 (0.08)	-0.13 (0.09)	-0.14 (0.09)	-0.25* (0.11)	-0.30* (0.12)
Gender: Male to Female	0.02 (0.10)	-0.01 (0.10)	-0.02 (0.09)	-0.02 (0.09)	-0.02 (0.08)	-0.07 (0.08)	0.03 (0.08)	0.02 (0.08)	-0.02 (0.09)	-0.06 (0.10)	-0.10 (0.11)
Gender: Female to Female	0.22 (0.15)	0.22 (0.13)	0.13 (0.14)	0.10 (0.12)	0.07 (0.12)	0.02 (0.12)	0.08 (0.11)	0.12 (0.10)	-0.04 (0.13)	0.02 (0.13)	-0.11 (0.17)
Party: Rep. to Dem.	0.78*** (0.19)	0.78*** (0.17)	0.81*** (0.17)	0.78*** (0.16)	0.63*** (0.17)	0.47** (0.15)	0.36* (0.16)	-0.01 (0.18)	0.09 (0.18)	-0.06 (0.22)	-0.06 (0.25)
Party: Dem. to Rep.	0.27 (0.19)	0.18 (0.17)	0.29 (0.17)	0.26 (0.16)	0.34* (0.17)	0.46** (0.15)	0.73*** (0.16)	0.88*** (0.18)	0.78*** (0.18)	1.09*** (0.22)	1.10*** (0.24)
Party: Rep. to Rep.	-0.11 (0.08)	-0.09 (0.07)	-0.04 (0.07)	-0.12 (0.07)	-0.18** (0.07)	-0.03 (0.06)	0.06 (0.07)	-0.00 (0.06)	0.15* (0.07)	0.31*** (0.08)	0.34*** (0.09)
Chamber: Senate to House	-1.68*** (0.20)	-1.77*** (0.19)	-1.88*** (0.20)	-1.67*** (0.17)	-1.75*** (0.18)	-1.81*** (0.18)	-1.62*** (0.18)	-1.53*** (0.17)	-1.33*** (0.16)	-1.19*** (0.17)	-1.06*** (0.18)
Chamber: House to Senate	-1.79*** (0.23)	-1.89*** (0.22)	-1.82*** (0.21)	-1.63*** (0.19)	-1.62*** (0.18)	-1.65*** (0.18)	-1.58*** (0.19)	-1.48*** (0.18)	-1.43*** (0.17)	-1.31*** (0.19)	-1.23*** (0.20)
Chamber: Senate to Senate	0.57*** (0.03)	0.52*** (0.03)	0.51*** (0.03)	0.50*** (0.02)	0.52*** (0.02)	0.50*** (0.02)	0.54*** (0.02)	0.49*** (0.03)	0.49*** (0.03)	0.43*** (0.05)	0.43*** (0.06)
State: Same	0.25* (0.10)	0.26** (0.10)	0.28** (0.09)	0.21* (0.09)	0.28*** (0.08)	0.30*** (0.08)	0.36*** (0.08)	0.37*** (0.07)	0.32*** (0.09)	0.35*** (0.09)	0.34*** (0.10)
Seniority: Difference	0.02*** (0.01)	0.02*** (0.01)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.02*** (0.00)	0.01 (0.01)	0.01 (0.01)	0.01 (0.01)
Ideology: Difference	-0.21 (0.22)	-0.29 (0.20)	-0.27 (0.19)	-0.29 (0.19)	-0.16 (0.19)	0.01 (0.17)	0.26 (0.18)	0.64*** (0.19)	0.55** (0.19)	0.82*** (0.23)	0.79** (0.25)
Num. Obs	285690	281430	286760	283556	284622	298662	283556	291060	292140	279312	283556
Log Likelihood	-117251.87	-132920.43	-146086.73	-157596.07	-165202.89	-166954.96	-157947.81	-152130.29	-141637.84	-127297.40	-112352.70
BIC	234679.62	266016.53	292349.39	315367.91	330581.60	334086.43	316071.40	304436.72	283451.86	254770.36	224881.17

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

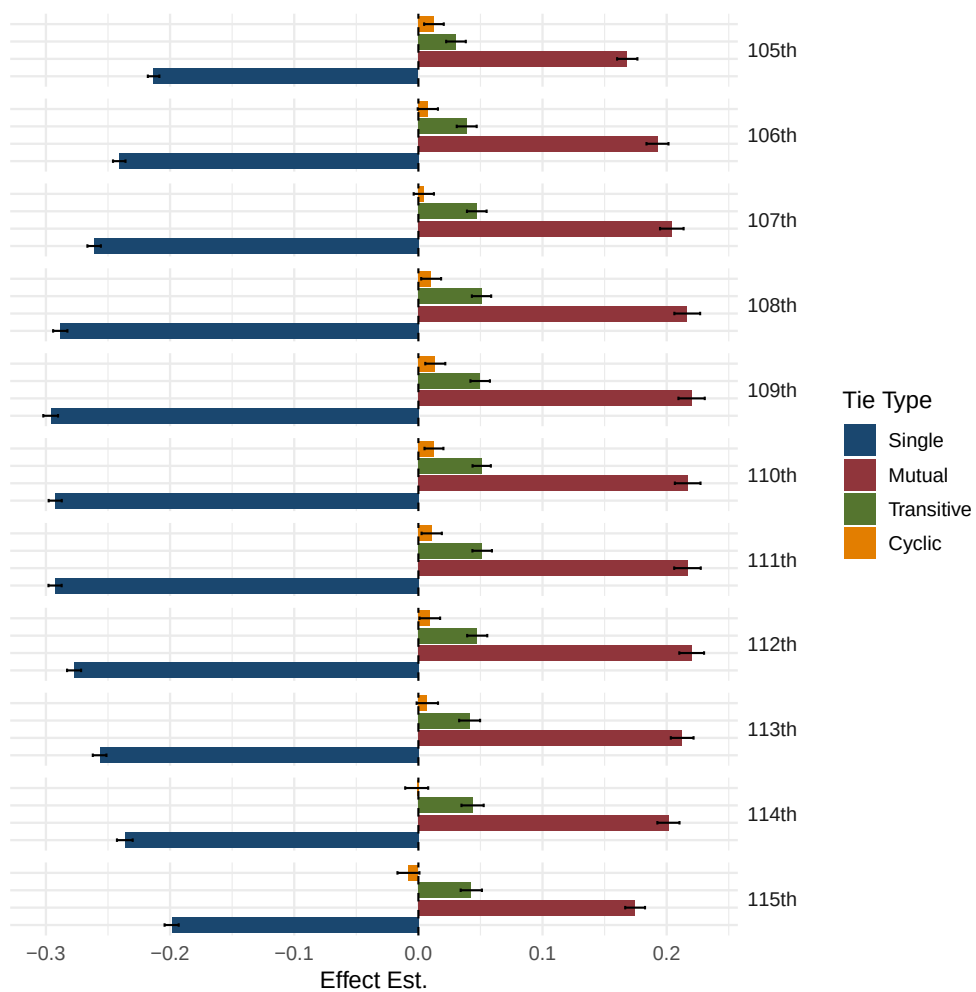


Figure A.7: Average marginal effects of tie types, all eleven Congresses.

Table A.17: All logistic fits (unweighted): 105th to 115th Congresses.

	105th	106th	107th	108th	109th	110th	111th	112th	113th	114th	115th
Intercept	-5.68*** (0.08)	-5.51*** (0.07)	-5.50*** (0.07)	-5.30*** (0.07)	-5.21*** (0.07)	-5.29*** (0.06)	-5.49*** (0.06)	-5.33*** (0.07)	-5.48*** (0.07)	-5.74*** (0.10)	-5.97*** (0.11)
Gender: Female to Male	0.14 (0.09)	0.11 (0.09)	0.00 (0.08)	0.02 (0.08)	-0.04 (0.08)	-0.19* (0.08)	-0.16* (0.08)	-0.13 (0.08)	-0.21* (0.08)	-0.36*** (0.10)	-0.47*** (0.11)
Gender: Male to Female	0.13 (0.09)	0.08 (0.09)	-0.01 (0.09)	0.01 (0.08)	-0.04 (0.08)	-0.18* (0.08)	-0.06 (0.08)	-0.06 (0.08)	-0.14 (0.08)	-0.25** (0.09)	-0.35*** (0.10)
Gender: Female to Female	0.49* (0.20)	0.44* (0.19)	0.22 (0.18)	0.19 (0.17)	0.10 (0.16)	-0.12 (0.16)	0.07 (0.15)	0.17 (0.14)	-0.17 (0.17)	-0.12 (0.18)	-0.34 (0.22)
Party: Rep. to Dem.	0.69*** (0.14)	0.69*** (0.13)	0.75*** (0.13)	0.66*** (0.12)	0.54*** (0.13)	0.56*** (0.11)	0.63*** (0.11)	0.30* (0.13)	0.44*** (0.13)	0.44** (0.17)	0.47* (0.18)
Party: Dem. to Rep.	0.43** (0.15)	0.39** (0.13)	0.49*** (0.13)	0.39** (0.12)	0.37** (0.13)	0.53*** (0.12)	0.81*** (0.12)	0.74*** (0.13)	0.77*** (0.13)	1.01*** (0.17)	1.03*** (0.19)
Party: Rep. to Rep.	-0.24* (0.10)	-0.19* (0.09)	-0.09 (0.09)	-0.26** (0.09)	-0.40*** (0.09)	-0.02 (0.08)	0.25** (0.09)	-0.04 (0.08)	0.27** (0.09)	0.48*** (0.11)	0.56*** (0.12)
Chamber: Senate to House	-2.19*** (0.20)	-2.33*** (0.19)	-2.47*** (0.20)	-2.26*** (0.17)	-2.31*** (0.18)	-2.40*** (0.18)	-2.22*** (0.18)	-2.20*** (0.17)	-2.00*** (0.16)	-1.86*** (0.17)	-1.72*** (0.18)
Chamber: House to Senate	-2.41*** (0.23)	-2.55*** (0.22)	-2.52*** (0.21)	-2.33*** (0.19)	-2.31*** (0.19)	-2.37*** (0.19)	-2.28*** (0.19)	-2.24*** (0.18)	-2.13*** (0.18)	-2.00*** (0.19)	-1.88*** (0.20)
Chamber: Senate to Senate	1.68*** (0.07)	1.49*** (0.06)	1.44*** (0.06)	1.47*** (0.06)	1.53*** (0.06)	1.49*** (0.06)	1.48*** (0.07)	1.16*** (0.07)	1.11*** (0.07)	0.72*** (0.10)	0.69*** (0.12)
State: Same	0.39** (0.13)	0.41*** (0.12)	0.45*** (0.12)	0.33** (0.12)	0.45*** (0.11)	0.48*** (0.11)	0.59*** (0.11)	0.62*** (0.11)	0.53*** (0.12)	0.57*** (0.13)	0.56*** (0.14)
Seniority: Difference	0.01* (0.00)	0.01* (0.00)	0.01** (0.00)	0.01** (0.00)	0.01** (0.00)	0.01** (0.00)	0.01* (0.00)	0.01** (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Ideology: Difference	-0.11 (0.16)	-0.15 (0.14)	-0.14 (0.14)	-0.16 (0.14)	-0.10 (0.14)	-0.01 (0.12)	0.12 (0.13)	0.31* (0.13)	0.26* (0.12)	0.40* (0.16)	0.38* (0.17)
Num. Obs.	285690	281430	286760	283556	284622	298662	283556	291060	292140	279312	283556
Log Likelihood	-7143.35	-7777.08	-8219.11	-8914.49	-9080.55	-9556.43	-8871.77	-8842.53	-8378.76	-6965.51	-5930.34
Pseudo R ²	0.08	0.08	0.08	0.08	0.08	0.08	0.07	0.06	0.05	0.04	0.04

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

Table A.18: All logistic fits (weighted): 105th to 115th Congresses.

	105th	106th	107th	108th	109th	110th	111th	112th	113th	114th	115th
Intercept	-5.58*** (0.08)	-5.42*** (0.08)	-5.41*** (0.08)	-5.22*** (0.07)	-5.12*** (0.07)	-5.21*** (0.06)	-5.43*** (0.06)	-5.27*** (0.07)	-5.45*** (0.08)	-5.74*** (0.10)	-5.97*** (0.11)
Gender: Female to Male	0.16 (0.10)	0.13 (0.10)	0.01 (0.09)	0.04 (0.08)	-0.03 (0.08)	-0.17* (0.08)	-0.10 (0.09)	-0.07 (0.08)	-0.13 (0.09)	-0.30** (0.11)	-0.42*** (0.12)
Gender: Male to Female	0.14 (0.10)	0.10 (0.10)	0.01 (0.09)	0.02 (0.08)	-0.03 (0.08)	-0.17* (0.08)	-0.04 (0.08)	-0.02 (0.08)	-0.09 (0.08)	-0.24* (0.10)	-0.35** (0.11)
Gender: Female to Female	0.48* (0.21)	0.43* (0.20)	0.23 (0.19)	0.17 (0.18)	0.11 (0.16)	-0.09 (0.16)	0.12 (0.16)	0.20 (0.15)	-0.13 (0.17)	-0.11 (0.18)	-0.32 (0.23)
Party: Rep. to Dem.	0.73*** (0.15)	0.75*** (0.13)	0.81*** (0.13)	0.72*** (0.13)	0.60*** (0.14)	0.61*** (0.12)	0.66*** (0.12)	0.33* (0.13)	0.48*** (0.13)	0.48** (0.18)	0.48* (0.19)
Party: Dem. to Rep.	0.52** (0.16)	0.49*** (0.14)	0.58*** (0.14)	0.49*** (0.14)	0.45** (0.14)	0.61*** (0.13)	0.90*** (0.13)	0.82*** (0.14)	0.86*** (0.14)	1.13*** (0.18)	1.16*** (0.20)
Party: Rep. to Rep.	-0.27* (0.11)	-0.21* (0.10)	-0.11 (0.10)	-0.28** (0.09)	-0.43*** (0.09)	-0.05 (0.09)	0.21* (0.09)	-0.08 (0.09)	0.24** (0.09)	0.49*** (0.11)	0.55*** (0.12)
Chamber: Senate to House	-2.21*** (0.21)	-2.35*** (0.21)	-2.47*** (0.22)	-2.30*** (0.19)	-2.34*** (0.19)	-2.38*** (0.19)	-2.20*** (0.19)	-2.16*** (0.18)	-1.98*** (0.18)	-1.81*** (0.19)	-1.66*** (0.20)
Chamber: House to Senate	-2.29*** (0.25)	-2.44*** (0.25)	-2.39*** (0.24)	-2.25*** (0.21)	-2.23*** (0.21)	-2.30*** (0.20)	-2.19*** (0.21)	-2.17*** (0.20)	-2.10*** (0.19)	-1.92*** (0.20)	-1.79*** (0.22)
Chamber: Senate to Senate	1.71*** (0.07)	1.53*** (0.07)	1.48*** (0.07)	1.50*** (0.06)	1.57*** (0.07)	1.55*** (0.06)	1.55*** (0.07)	1.24*** (0.08)	1.20*** (0.08)	0.90*** (0.12)	0.85*** (0.13)
State: Same	0.41** (0.14)	0.45*** (0.13)	0.48*** (0.13)	0.34** (0.13)	0.44*** (0.12)	0.47*** (0.11)	0.58*** (0.11)	0.61*** (0.11)	0.50*** (0.12)	0.54*** (0.13)	0.53*** (0.14)
Seniority: Difference	0.01* (0.00)	0.01* (0.00)	0.01** (0.00)	0.01** (0.00)	0.01** (0.00)	0.01** (0.00)	0.01** (0.00)	0.01** (0.00)	0.01 (0.00)	0.01 (0.00)	0.00 (0.01)
Ideology: Difference	-0.06 (0.18)	-0.11 (0.15)	-0.11 (0.15)	-0.12 (0.15)	-0.07 (0.16)	0.03 (0.14)	0.18 (0.14)	0.35* (0.14)	0.30* (0.13)	0.45** (0.17)	0.45* (0.18)
Num. Obs.	285690	281430	286760	283556	284622	298662	283556	291060	292140	279312	283556
Log Likelihood	-7143.35	-7777.08	-8219.11	-8914.49	-9080.55	-9556.43	-8871.77	-8842.53	-8378.76	-6965.51	-5930.34
Pseudo R ²	0.08	0.08	0.08	0.08	0.08	0.08	0.07	0.06	0.05	0.04	0.04

(1) *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.

(2) Edge weights used as regression weights in weighted logistic fits.

Table A.19: All zero-inflated Poisson fits: 105th to 115th Congresses.

	105th	106th	107th	108th	109th	110th	111th	112th	113th	114th	115th
Intercept	-2.38*** (0.11)	-2.27*** (0.10)	-2.29*** (0.10)	-2.25*** (0.09)	-2.21*** (0.09)	-2.36*** (0.09)	-2.54*** (0.09)	-2.19*** (0.10)	-2.40*** (0.11)	-2.58*** (0.14)	-2.70*** (0.16)
Gender: Female to Male	0.08 (0.10)	0.07 (0.09)	-0.01 (0.09)	0.03 (0.08)	-0.04 (0.08)	-0.16* (0.08)	-0.10 (0.08)	-0.07 (0.08)	-0.12 (0.09)	-0.31** (0.10)	-0.41*** (0.11)
Gender: Male to Female	0.06 (0.09)	0.04 (0.09)	-0.02 (0.09)	0.00 (0.08)	-0.04 (0.08)	-0.16* (0.08)	-0.03 (0.08)	-0.02 (0.08)	-0.07 (0.08)	-0.24** (0.09)	-0.34** (0.11)
Gender: Female to Female	0.15 (0.20)	0.16 (0.19)	0.04 (0.18)	-0.02 (0.17)	-0.03 (0.16)	-0.18 (0.16)	0.02 (0.15)	0.10 (0.14)	-0.13 (0.17)	-0.13 (0.18)	-0.31 (0.22)
Party: Rep. to Dem.	0.64*** (0.14)	0.68*** (0.13)	0.75*** (0.13)	0.66*** (0.13)	0.55*** (0.13)	0.60*** (0.12)	0.66*** (0.13)	0.33* (0.14)	0.50*** (0.14)	0.46** (0.18)	0.46* (0.20)
Party: Dem. to Rep.	0.53*** (0.14)	0.51*** (0.13)	0.58*** (0.13)	0.51*** (0.13)	0.47*** (0.13)	0.62*** (0.12)	0.89*** (0.13)	0.83*** (0.14)	0.90*** (0.14)	1.13*** (0.18)	1.17*** (0.20)
Party: Rep. to Rep.	-0.24* (0.10)	-0.17 (0.13)	-0.09 (0.09)	-0.26** (0.09)	-0.40*** (0.09)	-0.01 (0.08)	0.27** (0.09)	-0.05 (0.09)	0.30** (0.09)	0.49*** (0.11)	0.55*** (0.12)
Chamber: Senate to House	-2.22*** (0.19)	-2.35*** (0.18)	-2.47*** (0.19)	-2.29*** (0.17)	-2.33*** (0.17)	-2.37*** (0.17)	-2.19*** (0.17)	-2.15*** (0.16)	-1.96*** (0.15)	-1.80*** (0.16)	-1.65*** (0.17)
Chamber: House to Senate	-2.30*** (0.20)	-2.45*** (0.20)	-2.40*** (0.19)	-2.26*** (0.17)	-2.23*** (0.17)	-2.30*** (0.17)	-2.19*** (0.17)	-2.17*** (0.17)	-2.10*** (0.17)	-1.92*** (0.18)	-1.78*** (0.19)
Chamber: Senate to Senate	1.48*** (0.07)	1.34*** (0.07)	1.30*** (0.07)	1.33*** (0.07)	1.40*** (0.07)	1.42*** (0.06)	1.44*** (0.07)	1.16*** (0.08)	1.19*** (0.08)	0.91*** (0.10)	0.86*** (0.12)
State: Same	0.40** (0.13)	0.44*** (0.13)	0.46*** (0.12)	0.33** (0.12)	0.42*** (0.12)	0.45*** (0.11)	0.56*** (0.11)	0.56*** (0.11)	0.47*** (0.12)	0.51*** (0.13)	0.51*** (0.15)
Seniority: Difference	0.01* (0.00)	0.01* (0.00)	0.01* (0.00)	0.01** (0.00)	0.01** (0.00)	0.01* (0.00)	0.01* (0.00)	0.01* (0.00)	0.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Ideology: Difference	-0.01 (0.15)	-0.05 (0.14)	-0.07 (0.14)	-0.07 (0.13)	-0.04 (0.14)	0.04 (0.13)	0.17 (0.13)	0.36** (0.14)	0.30* (0.14)	0.45** (0.16)	0.46* (0.18)
Zero Inflation	3.10*** (0.07)	3.06*** (0.07)	3.04*** (0.07)	2.89*** (0.07)	2.84*** (0.07)	2.80*** (0.07)	2.84*** (0.07)	3.04*** (0.07)	3.05*** (0.08)	3.12*** (0.10)	3.23*** (0.12)
Num. Obs.	285690	281430	286760	283556	284622	298662	283556	291060	292140	279312	283556
Log Likelihood	-7834.12	-8499.32	-8966.00	-9706.80	-9885.73	-10326.23	-9545.32	-9467.61	-8879.52	-7291.44	-6178.71
BIC	15844.11	17174.30	18107.93	19589.38	19947.29	20828.96	19266.42	19111.36	17935.23	14758.44	12533.19

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$.